MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED LOW-POWER SCHOTTKY TTL, BINARY COUNTERS, CASCADABLE, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced low-power Schottky TTL, Binary Counters. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Synchronous 4-Bit Cascadable Binary Counter (Asynchronous Clear)
02	Synchronous 4-Bit Cascadable Binary Counter (Synchronous Clear)
03	Synchronous 4-Bit Cascadable UP/DOWN Binary Counter
04	Synchronous 4-Bit Cascadable Binary Counter with 3-State Outputs
05	Synchronous 4-Bit Cascadable UP/DOWN Binary Counter with 3-State Outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
2	C-2 (20 terminal, .350" x .350") square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1425) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device types 01 and 02	115 mW
Device type 03	137.5 mW
Device type 04	198 mW
Device type 05	176 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E, F, R, and S	(See MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1.4 Recommended operating conditions.

Maximum low level output current (I_{OL}):	
Device types 01, 02, and 03	4.0 mA
Device types 04 and 05	12.0 mA
Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL}) 2/	0.8 V dc
Normalized fanout (each output): 4/	
At low-logic level	10 maximum
At high-logic level	20 maximum
Input clock frequency (f_{MAX}):	
+25°C:	
Device types 01 and 02	40 MHz
Device types 03, 04, and 05	30 MHz
-55°C/+125°C:	
Device types 01 and 02	35 MHz
Device types 03, 04, and 05	25 MHz
Width of clock pulse (t_p clock):	
+25°C:	
Device types 01 and 02	12.5 ns
Device type 03	15 ns
-55°C/+125:	
Device types 01 and 02	14 ns
Device type 03	20 ns
+25°C:	
Device types 04 and 05	16.5 ns
-55°C/+125:	
Device types 04 and 05	20 ns
Width of Asynchronous clear pulse (t_p ACLR):	
Device types 01, 04, and 05	20 ns

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.
4/ Device will fanout in both high and low levels to the specified number of data data inputs on the same device type as that being tested.

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Width of asynchronous load pulse (t_{p} ALOAD):	
Device type 04	20 ns
Setup times before clock:	
DATA:	
Device types 01, 02, and 03	20 ns
Device types 04 and 05	25 ns
Synchronous clear:	
Low:	
Device types 02, 04, and 05	20 ns
Inactive:	
Device type 02	10 ns
Device types 04 and 05	35 ns
Asynchronous clear:	
Inactive:	
Device types 01, 04, and 05	10 ns
Asynchronous load:	
Inactive:	
Device type 04	10 ns
Synchronous load:	
Low:	
Device types 03, 04, and 05	20 ns
Inactive:	
Device types 04 and 05	35 ns
Enable P/Enable T:	
Low:	
Device types 01, 03, 04, and 05	25 ns
Device type 02	30 ns
High:	
Device types 01, 02, 03, and 04	25 ns
Device type 05	35 ns
Up/Down:	
Device type 03	20 ns
Device type 05	35 ns
Hold times (t_{HOLD}):	
Device types 01, 02, 03, 04, and 05	0 ns
Case operating temperature range (T_C)	-55°C to +125°C

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

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(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Typical operational sequence. The typical operational sequence shall be as specified on figure 4.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified		Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH1}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2.0 V	I_{OH} = -400 µA	01,02, 03	2.5		V
			RCO/CCO	04,05	2.5		
	V_{OH2}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2.0 V	I_{OH} = -1 mA	04,05	2.4		V
			Q outputs				
Low-level output voltage	V_{OL1}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2.0 V	I_{OL} = 4 mA All outputs	01,02, 03		0.4	V
			I_{OL} = 4 mA RCO/CCO	04,05		0.4	
	V_{OL2}		I_{OL} = 12 mA Q outputs	04,05		0.4	
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA	T_C = +25°C	All		-1.5	V
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		All	0	-200	µA
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	Load, CLK or ENT	01,02		40	µA
			ENT/ENP	04		40	
			All other inputs	01,02 03,04, 05		20	
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	Load, CLK or ENT	01,02		200	µA
			ENT/ENP	04		200	
			All other inputs	01,02, 03,04, 05		100	
Short circuit output	I_O 1/	V_{CC} = 5.5 V, V_O = 2.25 V	Q outputs	All	-30	-112	mA
			RCO	01,02,03	-30	-112	
			RCO and CCO	04,05	-15	-70	
Output current, low level outputs OFF	I_{OZL}	V_{CC} = 5.5 V, V_O = 0.4 V	Q outputs	04,05		-20	µA
Output current, high level outputs OFF	I_{OZH}	V_{CC} = 5.5 V, V_O = 2.7 V	Q outputs	04,05		20	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Supply current	I_{CC}	V_{CC} = 5.5 V	2/ 01, 02		25	mA
			03		25	
Supply current, high level	I_{CCH}	V_{CC} = 5.5 V	04		27	mA
			05		26	
Supply current, low level	I_{CCL}	V_{CC} = 5.5 V	04		33	mA
			05		32	
Supply current, outputs disabled	I_{CCZ}	V_{CC} = 5.5 V	04		35	mA
			05		32	
Maximum input A, clock, or count up frequency	f_{MAX}		All	25		MHz
Propagation delay time low-to-high, CLK-Q	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 500Ω	01,02	4	18	ns
			03	5	19	
			04	4	15	
			05	4	17	
Propagation delay time high-to-low, CLK-Q	t_{PHL1}		01,02	6	25	ns
			03	5	20	
			04	5	21	
			05	7	18	
Propagation delay time low-to-high, CLK-RCO	t_{PLH2}		01,02	5	25	ns
			03	6	25	
			04	9	35	
			05	12	31	
Propagation delay time high-to-low, CLK-RCO	t_{PHL2}		01,02	5	25	ns
			03	6	25	
			04	8	29	
			05	10	22	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Propagation delay time low-to-high, CLK-CCO	t_{PLH3}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 500Ω	04	8	31	ns
			05	5	15	
Propagation delay time high-to-low, CLK-CCO	t_{PHL3}		04	5	20	ns
			05	6	30	
Propagation delay time low-to-high, ALOAD-Q	t_{PLH4}		04	10	38	ns
Propagation delay time high-to-low, ALOAD-Q	t_{PHL4}		04	7	27	ns
Propagation delay time low-to-high, ALOAD-RCO	t_{PLH5}		04	15	55	ns
Propagation delay time high-to-low, ALOAD-RCO	t_{PHL5}		04	12	35	ns
Propagation delay time low-to-high, ALOAD-CCO	t_{PLH6}		04	25	65	ns
Propagation delay time high-to-low, ALOAD-CCO	t_{PHL6}		04	12	42	ns
Propagation delay time low-to-high, DATA-Q	t_{PLH7}		04	8	35	ns
Propagation delay time high-to-low, DATA-Q	t_{PHL7}		04	7	27	ns
Propagation delay time low-to-high, ENT-RCO	t_{PLH8}		01,02	3	16	ns
			04	5	20	
			03	5	19	
			05	6	17	
Propagation delay time high-to-low, ENT-RCO	t_{PHL8}		01,02	3	16	ns
			03	3	19	
			04	4	18	
			05	4	17	
Propagation delay time low-to-high, ENT-CCO	t_{PLH9}		04	12	35	ns
			05	5	15	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Propagation delay time high-to-low, ENT-CCO	t_{PHL9}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 500Ω	04	4	15	ns
			05	9	28	
Propagation delay time low-to-high, ENP-CCO	t_{PLH10}		04	5	22	ns
			05	4	14	
Propagation delay time high-to-low, ENP-CCO	t_{PHL10}		04	4	14	ns
			05	5	17	
Propagation delay time low-to-high, U/D-RCO	t_{PLH11}		03	5	25	ns
			05	9	25	
Propagation delay time high-to-low, U/D-RCO	t_{PHL11}		03	5	25	ns
			05	9	23	
Propagation delay time high-to-low, CLR-Q	t_{PHL12}		01	8	27	ns
			04	7	28	
			05	9	22	
Propagation delay time high-to-low, CLR-RCO	t_{PHL13}		01	11	28	ns
Enable time to high-level output	t_{PZH}	V_{CC} = 5.0 V C_L = 50 pF ±10% $R_1 = R_L$ = 500Ω	04	5	24	ns
			05	6	21	
Enable time to low-level output	t_{PZL}		04	8	28	ns
			05	11	29	
Disable time from high-level	t_{PHZ}		04	2	15	ns
			05	1	8	
Disable time from low-level	t_{PLZ}		04	4	20	ns
			05	3	19	

1/ The output conditions have been chosen to produce a current that closely approximates one-half of the true short-circuit output current, I_{OS}.

2/ I_{CC} measured: (a) With the clock input low, all other inputs high and all outputs open and (b) Then again with the clock input low with all other inputs low and all outputs open.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,9,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	---
Group C end-point electrical parameters (method 5005)	---	1,2,3
Additional electrical subgroups for group C periodic inspections	---	None
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

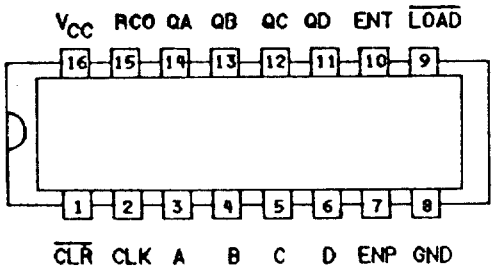
a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

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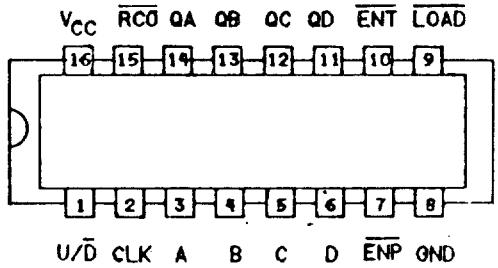
Device type 01 and 02

Cases E and F



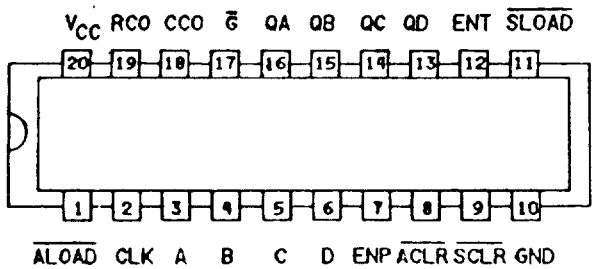
Device type 03

Cases E and F



Device type 04

Cases R and S



Device type 05

Cases R and S

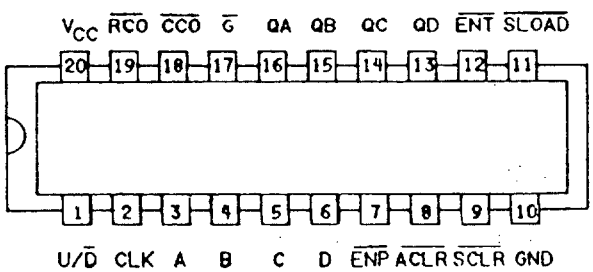
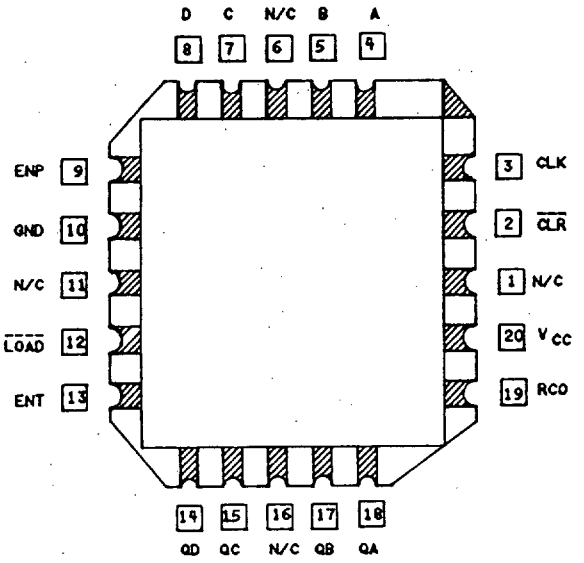


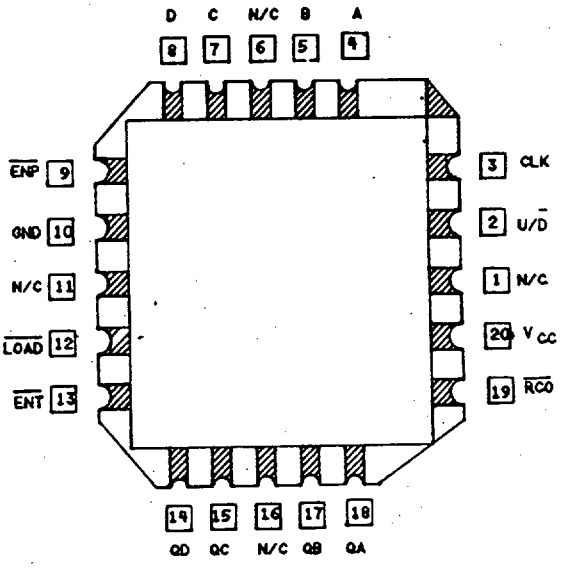
FIGURE 1. Terminal connections (top view).

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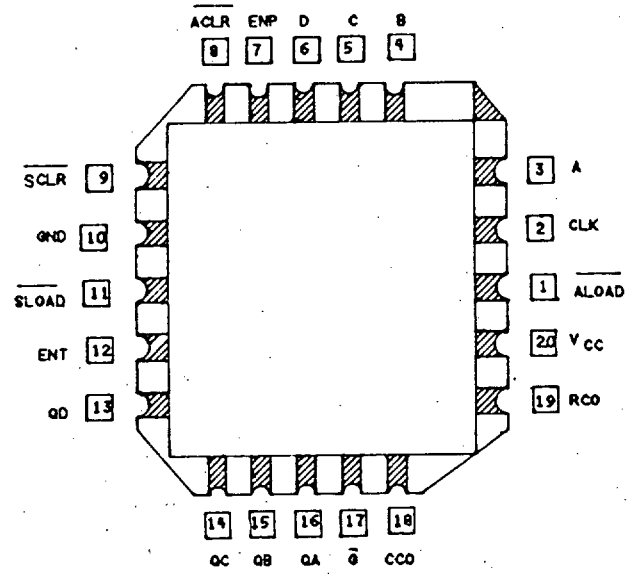
Device types 01 and 02
Case 2



Device type 03
Case 2



Device type 04
Case 2



Device type 05
Case 2

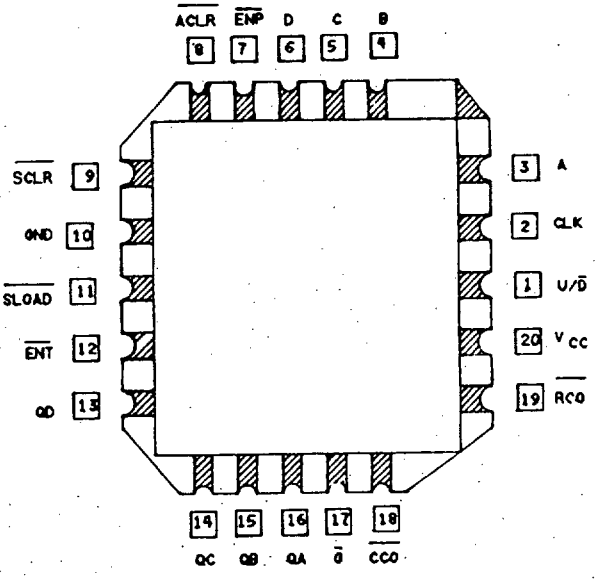


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01

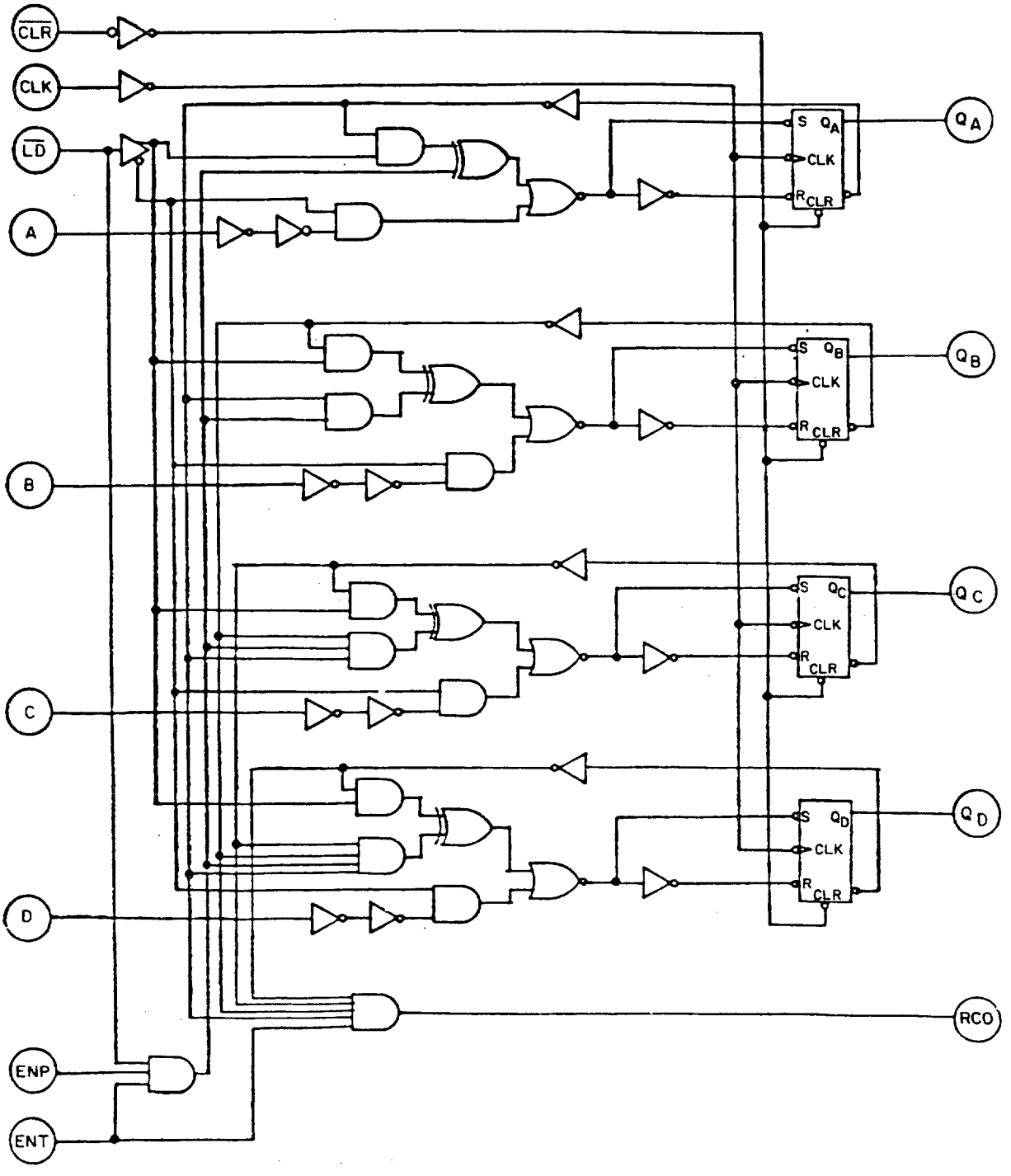
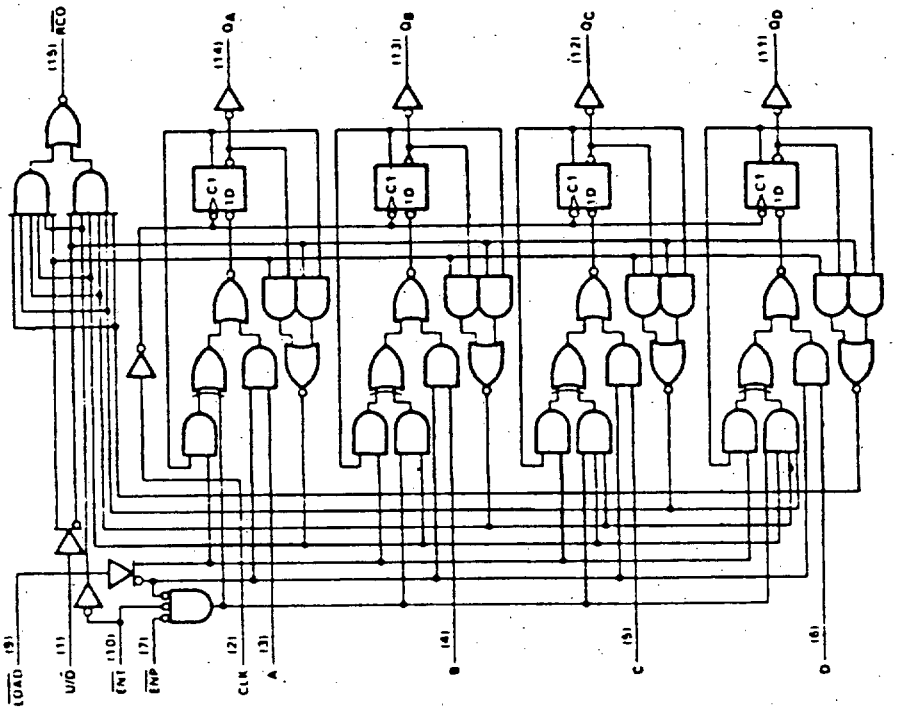
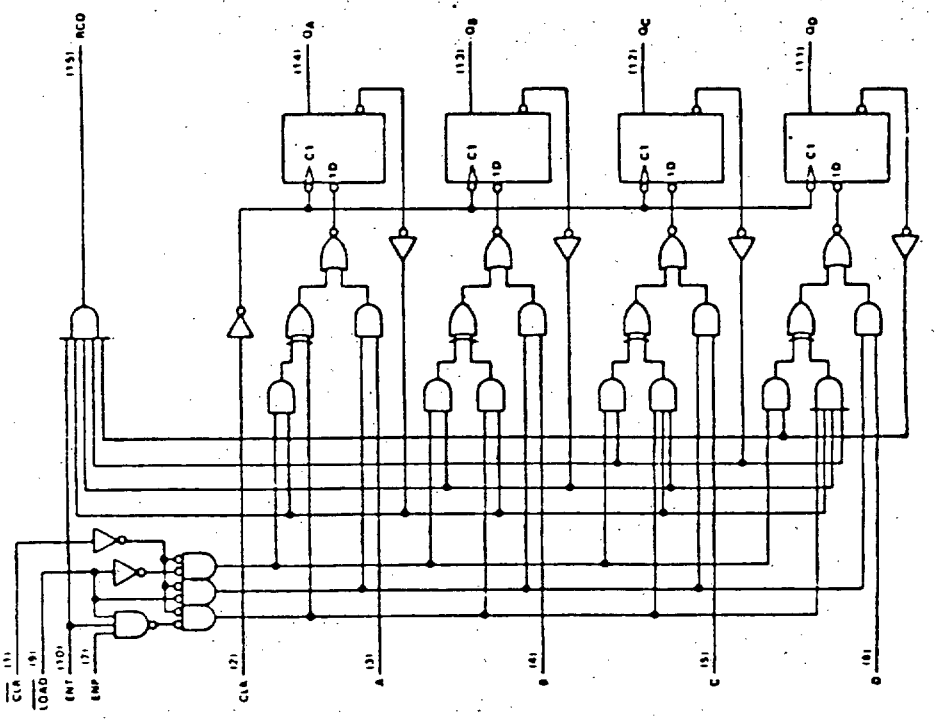


FIGURE 2. Logic diagrams .

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Device type 03



Device type 02

FIGURE 2. Logic diagrams - Continued.

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Device type 04

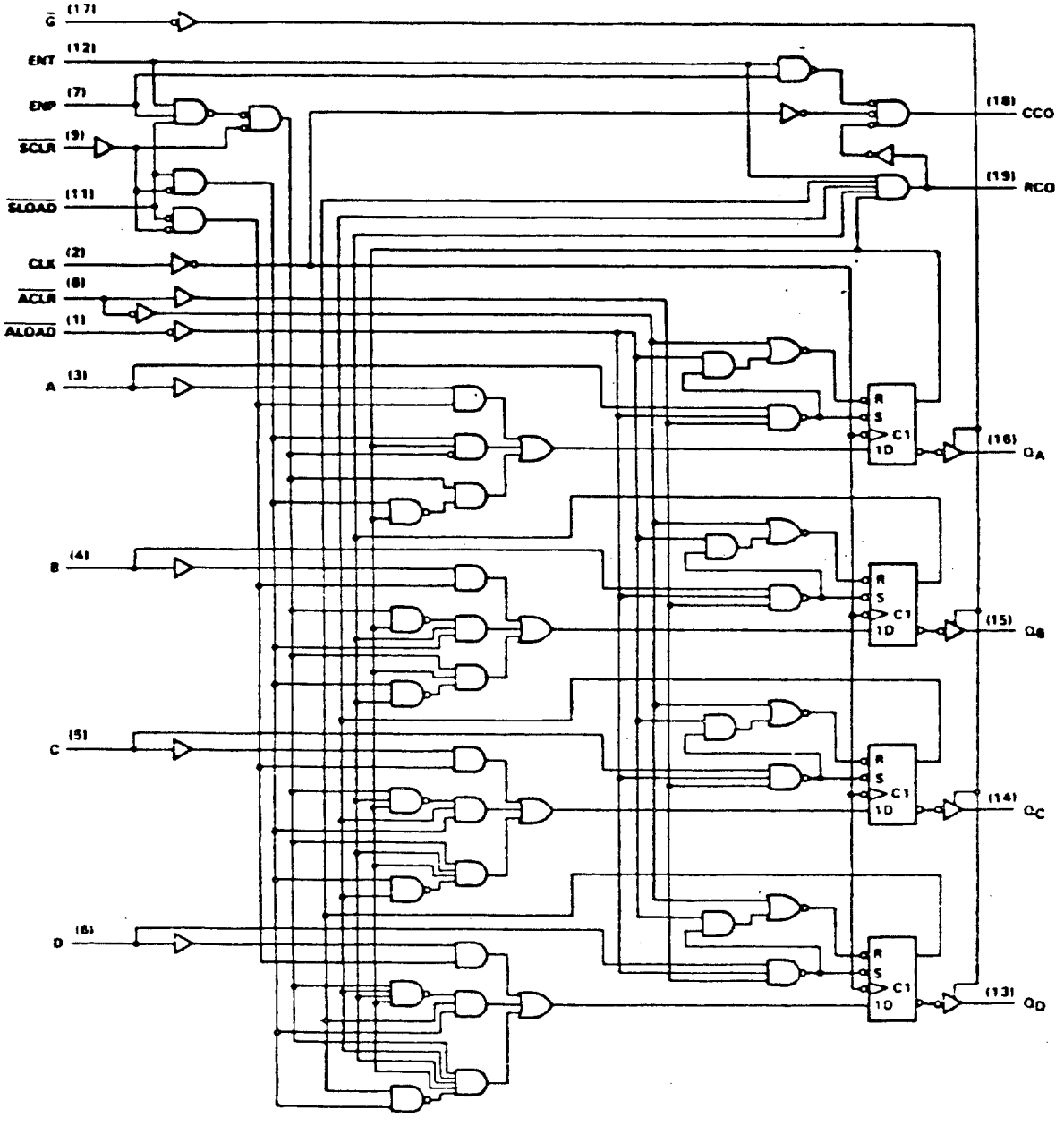


FIGURE 2. Logic diagrams - Continued.

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Device type 05

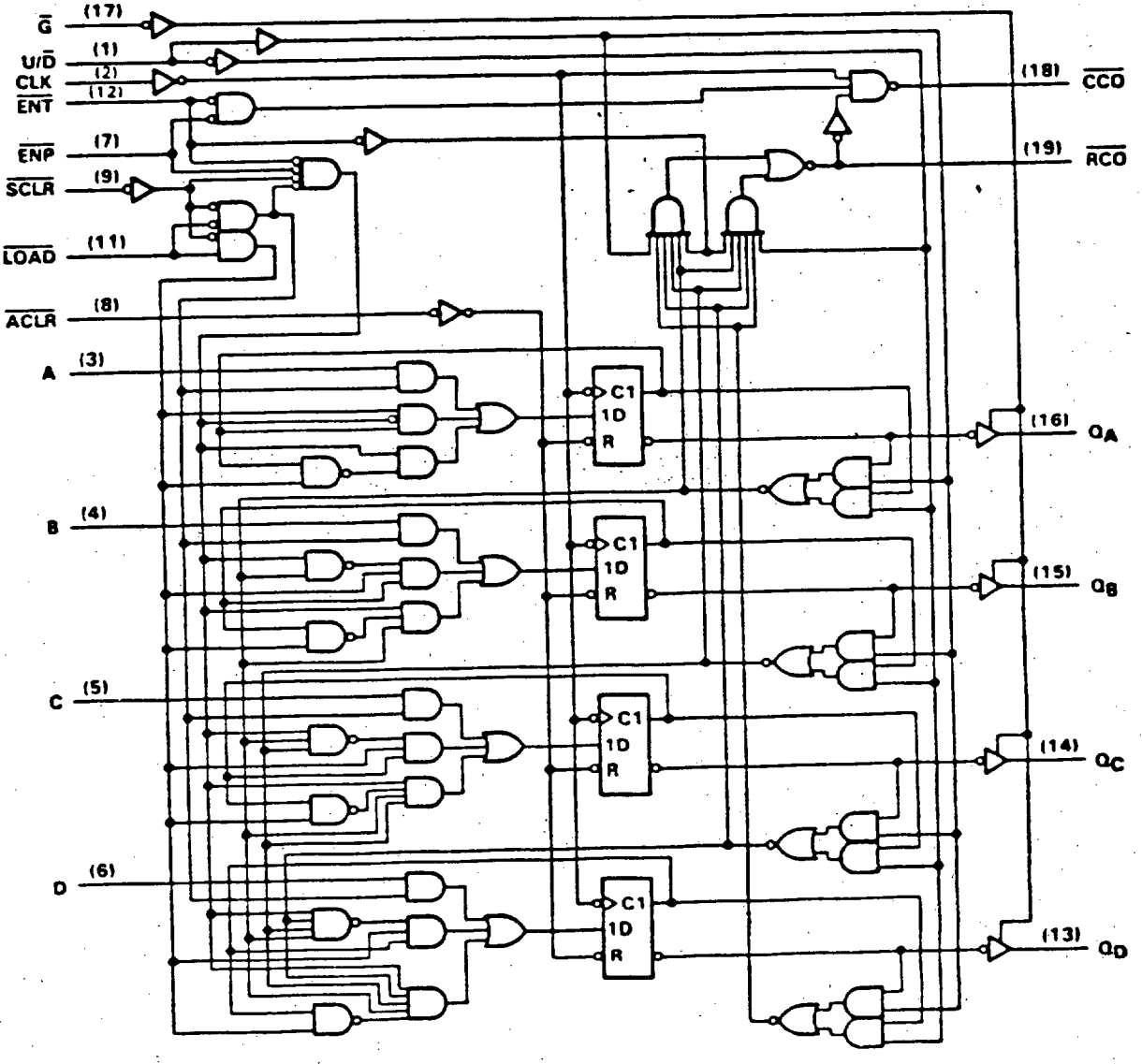


FIGURE 2. Logic diagrams - Continued.

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SYNCHRONOUS TRUTH TABLE, DEVICE TYPES 01 and 02

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output (RCO)
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (note 1)				H if count = 15 L if count < 15
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 15 L if count < 15
CP	X	L	L	X	X	X	X	H	A	B	C	D	L
CP	X	X	X	X	X	X	X	L	L	L	L	L	L

ASYNCHRONOUS TRUTH TABLE, DEVICE TYPE 01

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
X	X	X	X	X	X	X	X	L	L	L	L	L	L

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	H

NOTES:
1. See up count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = Clock pulse.
6. NC = No change.
7. The last line of synchronous truth table does not apply to device type 01.

FIGURE 3. Truth tables.

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Device type 03

Operation	Inputs at time t_n									Outputs at time t_{n+1}				
	Clock	ENP	ENT	Load	A	B	C	D	U/D	Q_A	Q_B	Q_C	Q_D	RCO
Load	↑	L	L	L	X	X	X	X	X	A	B	C	D	L if count=15 H if count≠15
Count Up	↑	L	L	H	X	X	X	X	H	Previous count plus 1 (note 1)				L if count=15 H if count≠15
Count Down	↑	L	L	H	X	X	X	X	L	Previous count minus 1 (note 2)				L if count=0 H if count≠0
Inhibit	↑	H	L	H	X	X	X	X	X	No change				No change
	↑	L	H	H	X	X	X	X	X					L
	↑	H	H	H	X	X	X	X	X					L

COUNT UP SEQUENCE				
Q_A	Q_B	Q_C	Q_D	RCO
L	L	L	L	H
L	L	L	H	H
L	L	H	L	H
L	L	H	H	H
L	H	L	L	H
L	H	L	H	H
L	H	H	L	H
L	H	H	H	H
H	L	L	L	H
H	L	L	H	H
H	L	H	L	H
H	L	H	H	H
H	H	L	L	H
H	H	L	H	H
H	H	H	L	H
H	H	H	H	L

COUNT DOWN SEQUENCE				
Q_A	Q_B	Q_C	Q_D	RCO
H	H	H	H	H
H	H	H	L	H
H	H	L	H	H
H	H	L	L	H
H	L	H	H	H
H	L	H	L	H
H	L	L	H	H
H	L	L	L	H
L	H	H	H	H
L	H	H	L	H
L	H	L	H	H
L	H	L	L	H
L	L	H	H	H
L	L	H	L	H
L	L	L	H	H
L	L	L	L	L

FIGURE 3. Truth tables - Continued.

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Device type 04

OPERATION	Inputs at time t_n												Outputs at time t_{n+1}					
	G	ACLR	ALOAD	SCLR	SLOAD	ENT	ENP	CLK	A	B	C	D	Q_A	Q_B	Q_C	Q_D	RCO	CCO
Q OUTPUTS DISABLED 3/	H	X	X	X	X	X	X	X	X	X	X	X	Z	Z	Z	Z	H if cnt=15 L if cnt≠15	note 1/
ASYNCHRONOUS CLEAR	L	L	X	X	X	X	X	X	X	X	X	X	L	L	L	L	L	L
ASYNCHRONOUS LOAD	L	H	L	X	X	X	X	X	X	X	X	X	A	B	C	D	H if cnt=15 L if cnt≠15	note 1/
SYNCHRONOUS CLEAR	L	H	H	L	X	X	X	↑	X	X	X	X	L	L	L	L	L	L
SYNCHRONOUS LOAD	L	H	H	H	L	X	X	↑	X	X	X	X	A	B	C	D	H if cnt=15 L if cnt≠15	note 1/
COUNT	L	H	H	H	H	H	H	↑	X	X	X	X	Previous count plus 1				H if cnt=15 L if cnt≠15	note 1/
INIBIT COUNTING	L	H	H	H	H	X	L	X	X	X	X	X	No Change				H if cnt=15 L if cnt≠15	note 1/
	L	H	H	H	H	L	X	X	X	X	X	X						
	L	H	H	H	H	L	L	X	X	X	X	X						

1/ CCO produces a high level pulse for a duration equal to that of the low level of the clock when RCO is high and the counter is enabled, otherwise CCO is high.

COUNT UP SEQUENCE					
Q_A	Q_B	Q_C	Q_D	RCO	CCO
L	L	L	L	L	L
L	L	L	H	L	L
L	L	H	L	L	L
L	L	H	H	L	L
L	H	L	L	L	L
L	H	L	H	L	L
L	H	H	L	L	L
L	H	H	H	L	L
H	L	L	L	L	L
H	L	L	H	L	L
H	L	H	L	L	L
H	L	H	H	L	L
H	H	L	L	L	L
H	H	L	H	L	L
H	H	H	L	L	L
H	H	H	H	H	1/

FIGURE 3. Truth tables - Continued.

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Device type 05

OPERATION	Input at time t_n												Outputs at time t_{n+1}					
	G	ACLR	SCLR	LOAD	ENT	ENP	U/D	CLK	A	B	C	D	Q_A	Q_B	Q_C	Q_D	RCO	CCO
Q OUTPUTS DISABLED 3/	H	X	X	X	X	X	X	X	X	X	X	X	Z	Z	Z	Z	L if cnt=15 H if cnt≠15	note 1/
ASYNCHRONOUS CLEAR	L	L	X	X	X	X	X	X	X	X	X	X	L	L	L	L	H	H
SYNCHRONOUS CLEAR	L	H	L	X	X	X	X	↑	X	X	X	X	L	L	L	L	H	H
LOAD	L	H	H	L	X	X	X	↑	X	X	X	X	A	B	C	D	L if cnt=15 H if cnt≠15	note 1/
COUNT UP	L	H	H	H	L	L	H	↑	X	X	X	X	Previous count plus 1				L if cnt=15 H if cnt≠15	note 1/
COUNT DOWN	L	H	H	H	L	L	L	↑	X	X	X	X	Previous count minus 1				L if cnt=0 H if cnt≠0	note 1/
INHIBIT COUNT	L	H	H	H	X	H	X	X	X	X	X	X	No Change (hold)				note 2/	note 1/
	L	H	H	H	H	X	X	X	X	X	X	X						
	L	H	H	H	H	H	X	X	X	X	X	X						

NOTES:
1/ CCO produces a low level pulse for the duration equal to that of the low level of the clock when RCO is low and the counter is enabled; otherwise CCO is high.
2/ RCO produces a low level pulse while the count is 15 when counting up, or while the count is 0 when counting down.
3/ Counting continues.

COUNT UP SEQUENCE					
Q_A	Q_B	Q_C	Q_D	RCO	CCO
L	L	L	L	H	H
L	L	L	H	H	H
L	L	H	L	H	H
L	L	H	H	H	H
L	H	L	L	H	H
L	H	L	H	H	H
L	H	H	L	H	H
L	H	H	H	H	H
H	L	L	L	H	H
H	L	L	H	H	H
H	L	H	L	H	H
H	L	H	H	H	H
H	H	L	L	H	H
H	H	L	H	H	H
H	H	H	L	H	H
H	H	H	H	L	1/

COUNT DOWN SEQUENCE					
Q_A	Q_B	Q_C	Q_D	RCO	CCO
H	H	H	H	H	H
H	H	H	L	H	H
H	H	L	H	H	H
H	H	L	L	H	H
H	L	H	H	H	H
H	L	H	L	H	H
H	L	L	H	H	H
H	L	L	L	H	H
L	H	H	H	H	H
L	H	H	L	H	H
L	H	L	H	H	H
L	H	L	L	H	H
L	L	H	H	H	H
L	L	H	L	H	H
L	L	L	H	H	H
L	L	L	L	L	1/

FIGURE 3. Truth tables - Continued.

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Device types 01 and 02

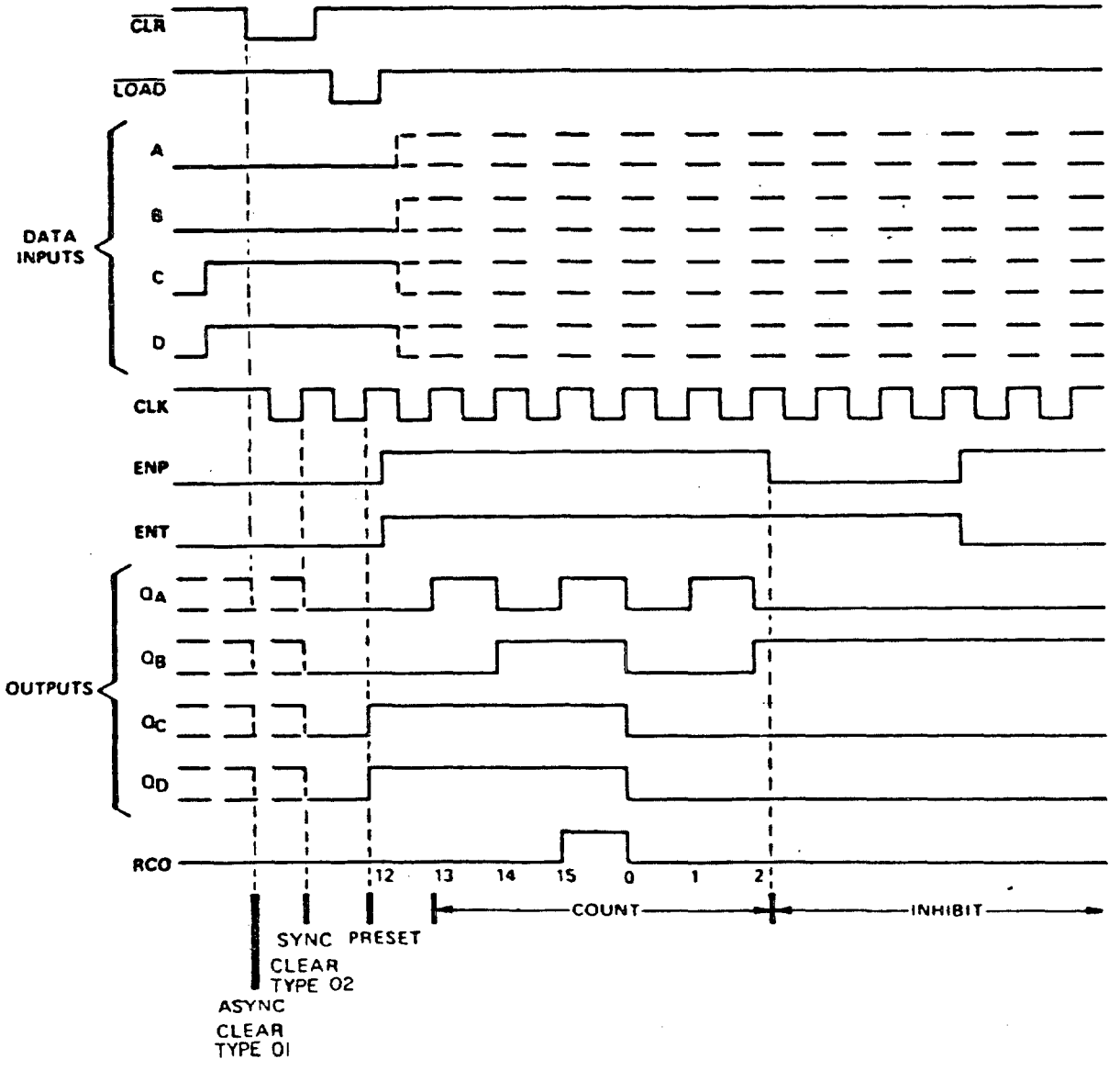


FIGURE 4. Typical operational sequence.

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Device type 03

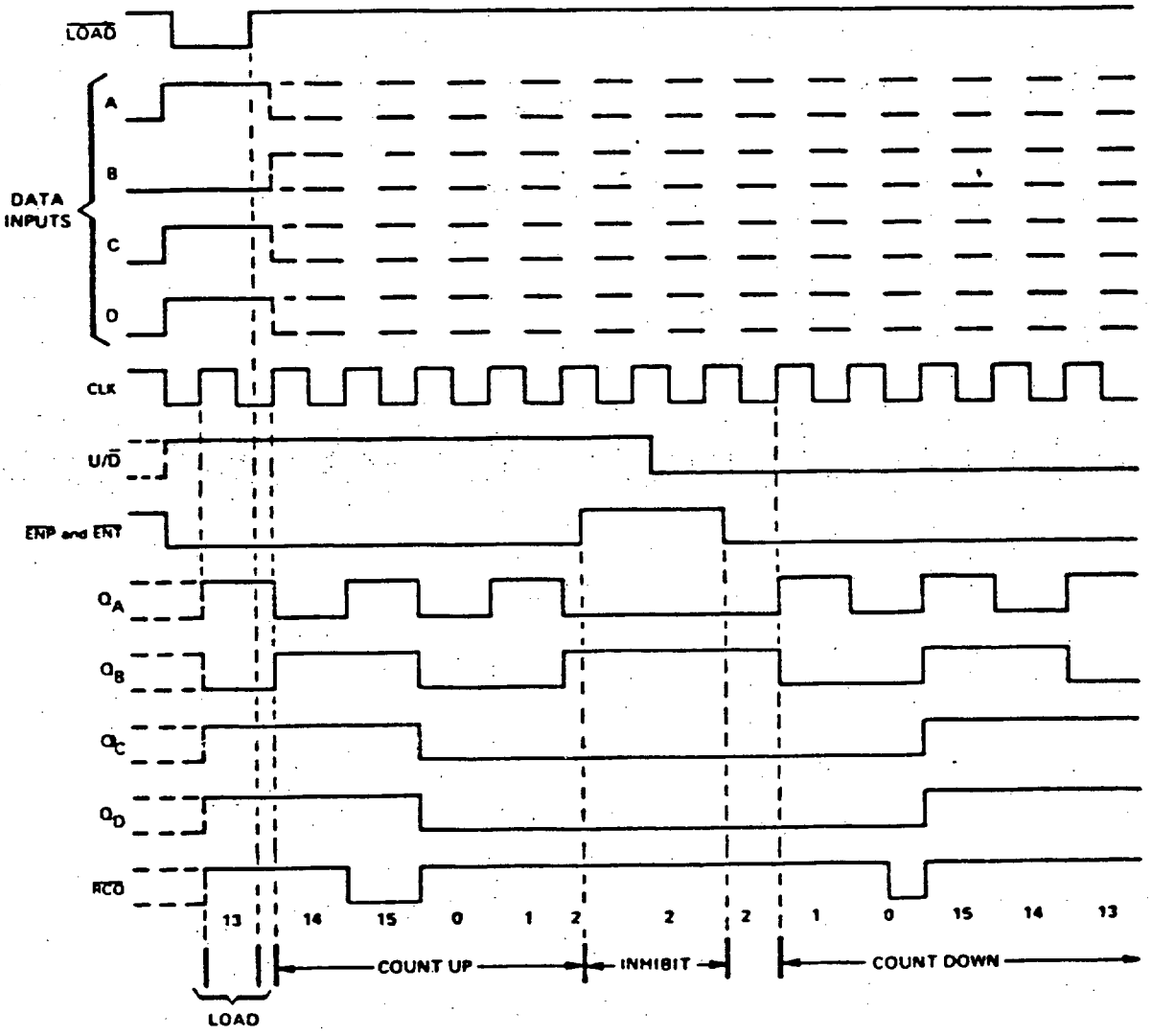


FIGURE 4. Typical operational sequence - Continued.

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Device type 04

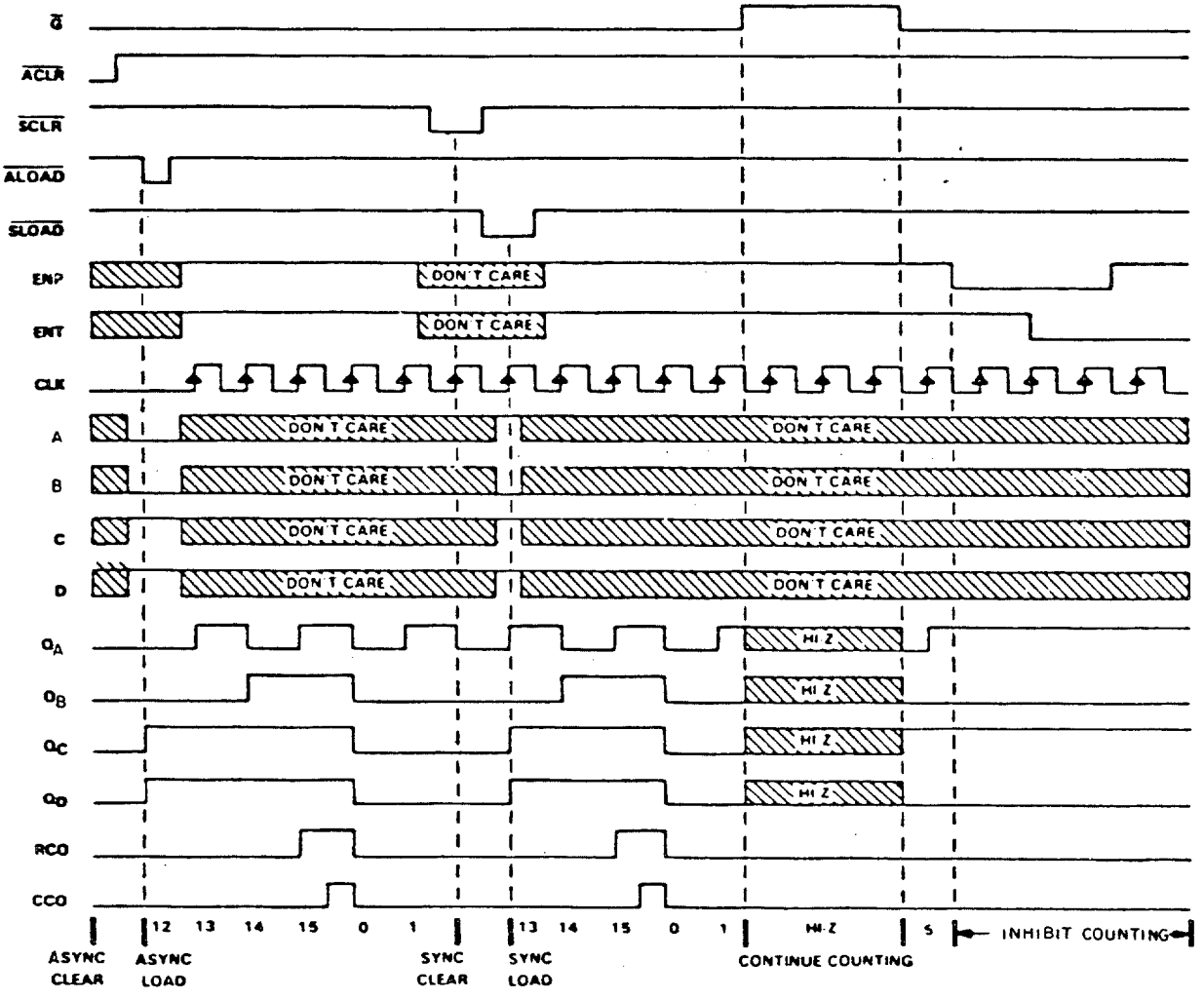


FIGURE 4. Typical operational sequence - Continued.

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Device type 05

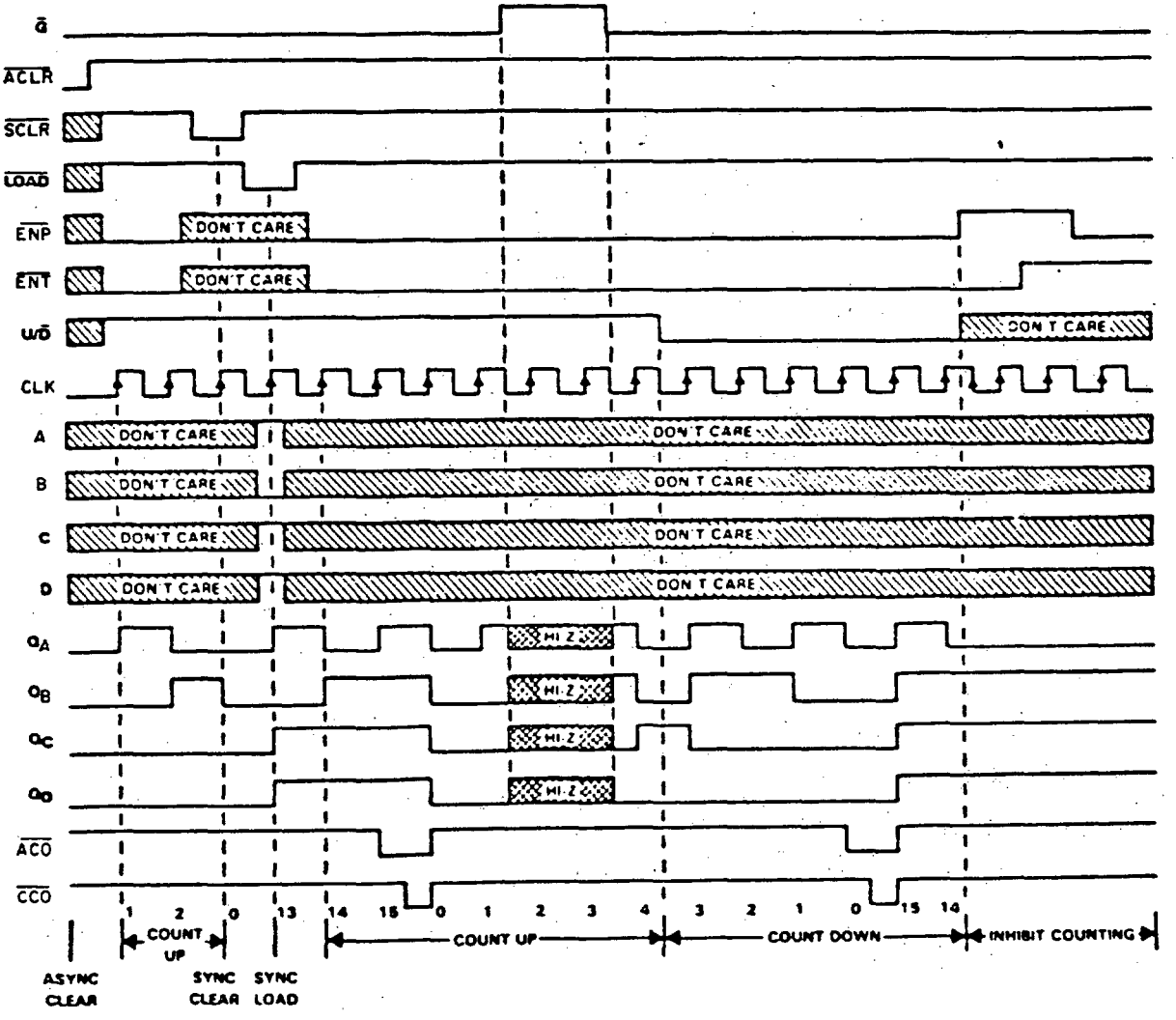
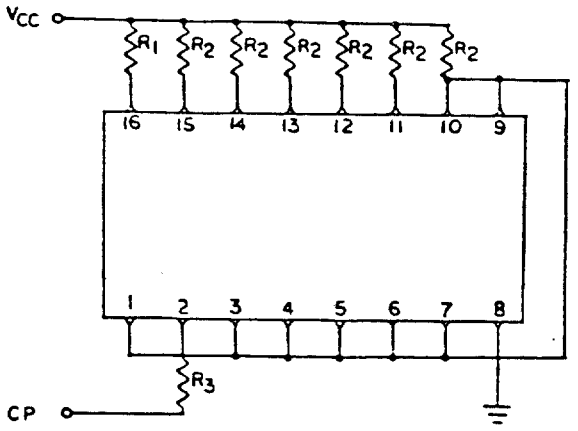


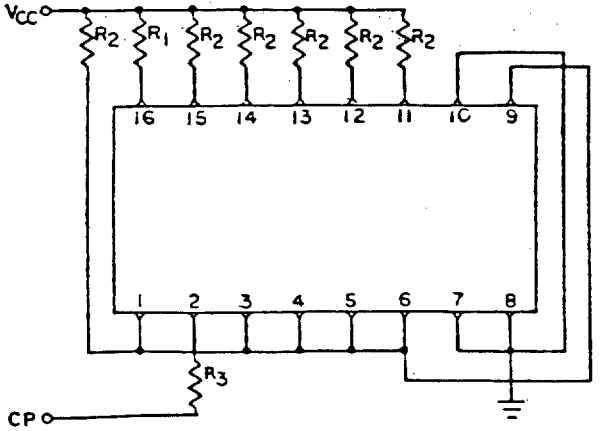
FIGURE 4. Typical operational sequence - Continued.

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Device types 01 and 02



Device type 03

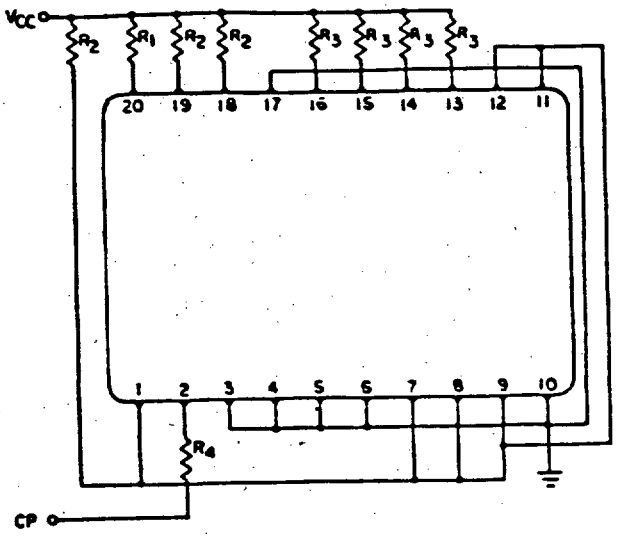


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50% ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum, V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R_2 = 1 kΩ ±5%; R_3 = 27Ω maximum.
3. V_{CC} and R_1 shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal.

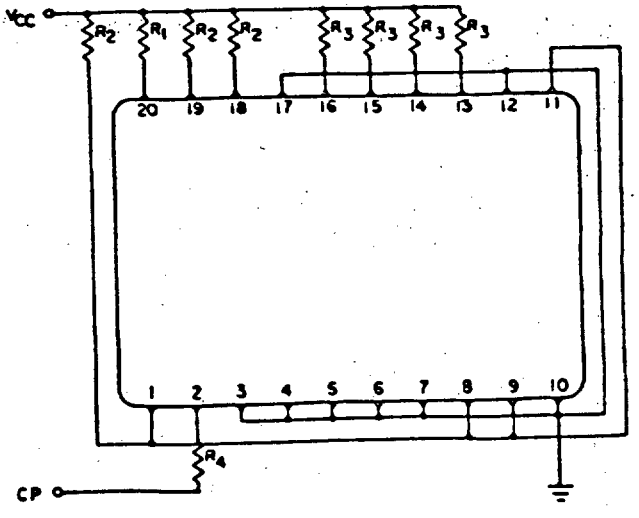
FIGURE 5. Burn-in and life test circuit.

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Device type 04



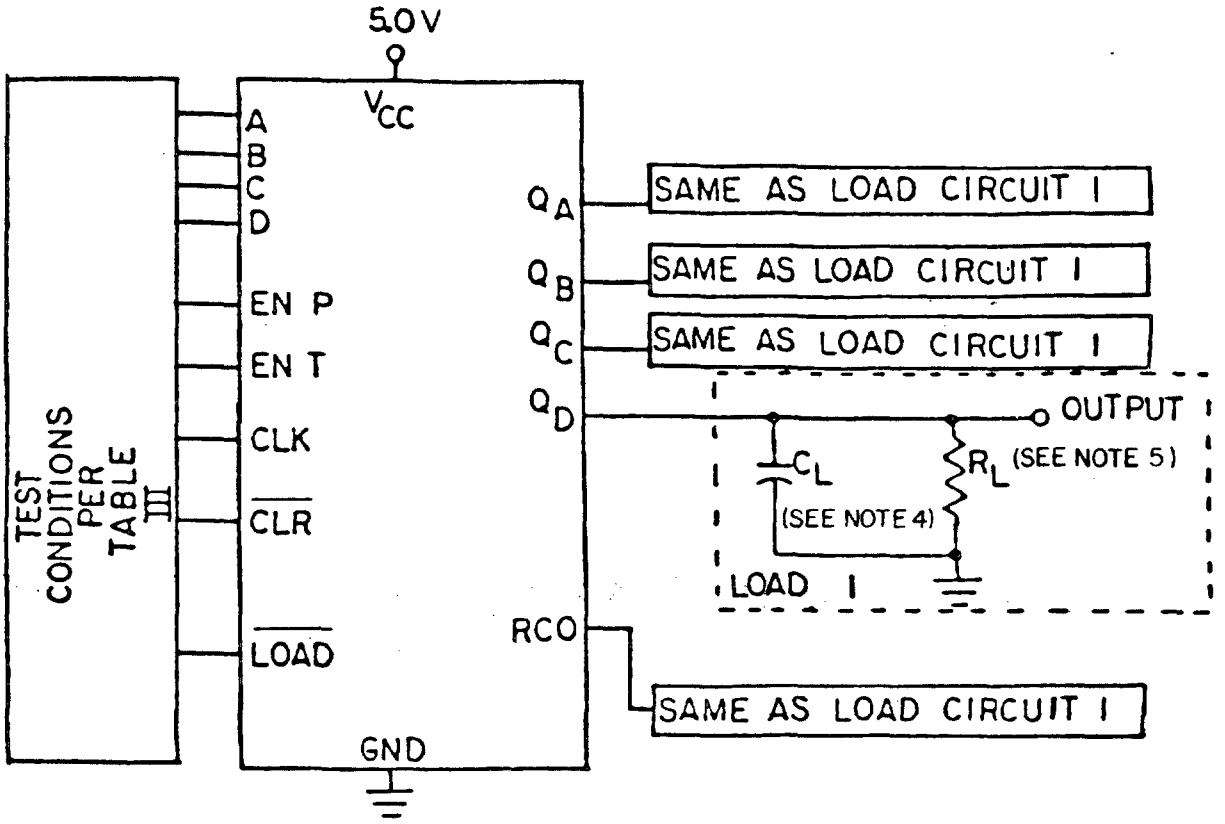
Device type 05



NOTES:
1. R_2 = 1 kΩ ±5%; R_3 = 470Ω ±5%; R_4 = 27Ω maximum.
2. CP = 100 kHz ±50% square wave; duty cycle = 50% ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum, V_{IH} = 2.0 V minimum to 5.5 V maximum.
3. V_{CC} and R_1 shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal.

FIGURE 5. Burn-in and life test circuit - Continued.

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NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 14 ns; PRR $\leq$ 1 MHz.
2. Clear input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clear) = 20 ns.
3. For f_{MAX}, the clock input pulse are as follows: $t_1 = t_0 = 3 \pm 1.5$ ns; for 25°C, t_p (clock) = 12.5 ns; PRR = 40 MHz; for -55/125°C, t_p (clock) = 14.5 ns; PRR = 35 MHz.
4. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
5. R_L = 499Ω±1%.
6. Voltage measurements are to be made with respect to network ground terminal.
7. Data and load inputs may be used to preset the device to the required count prior to test.

FIGURE 6. Switching time test circuit and waveforms for device types 01 and 02.

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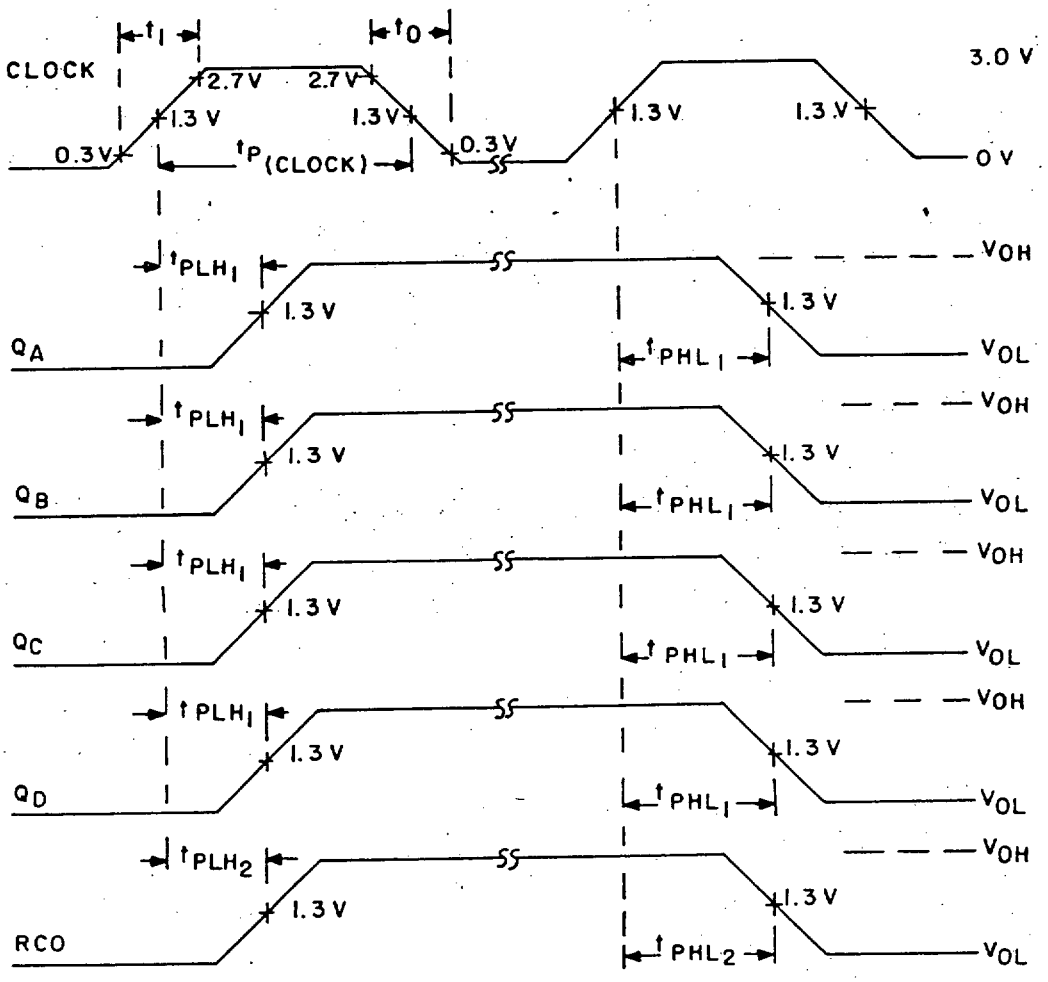
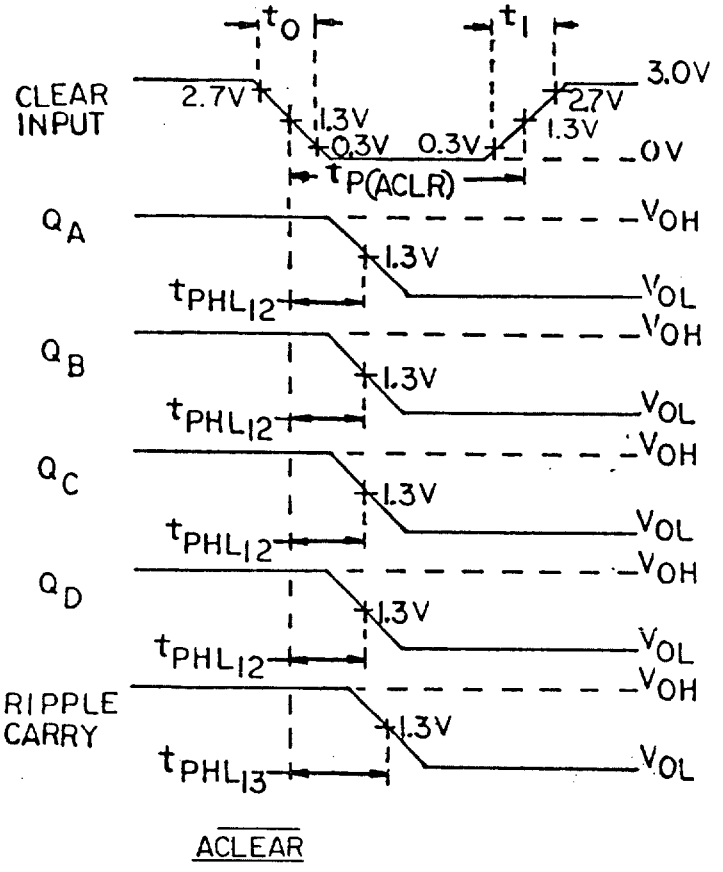


FIGURE 6. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

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Device type 01



Device types 01 and 02

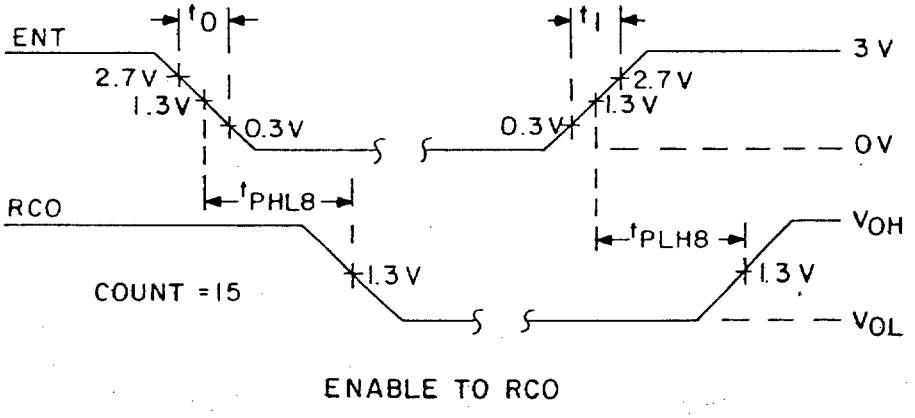
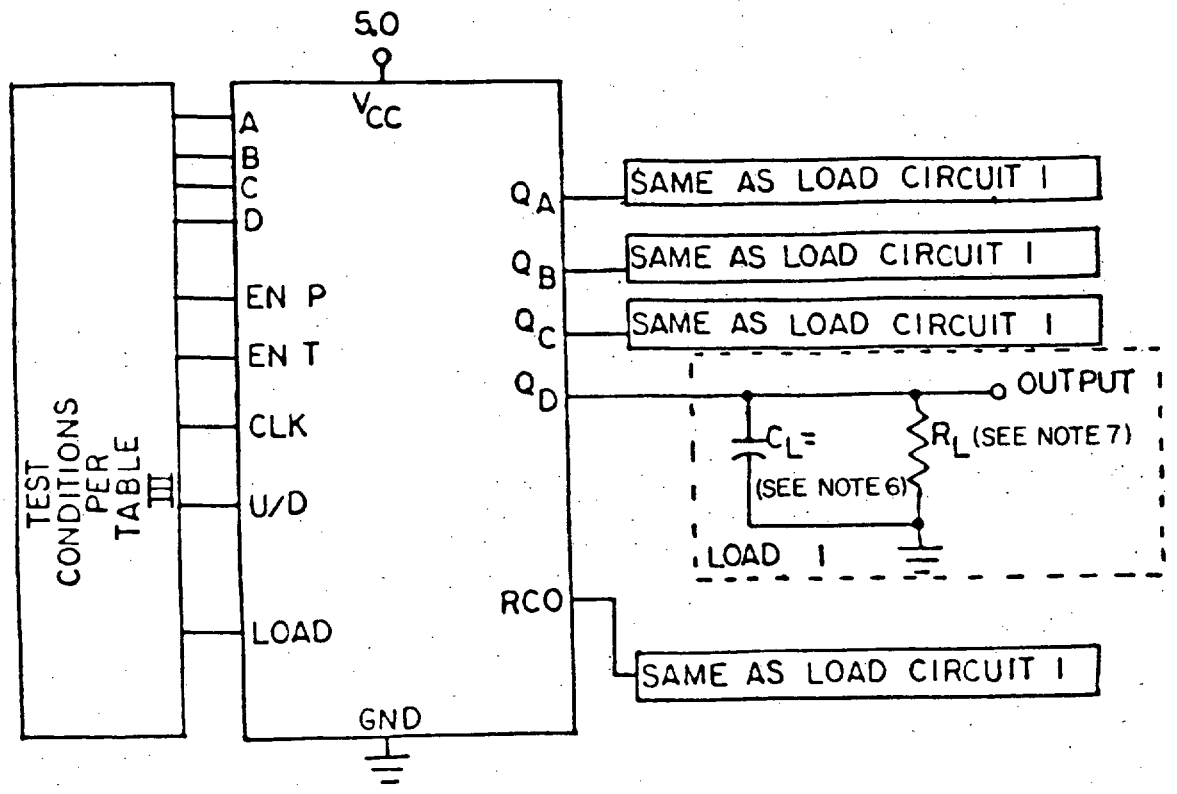


FIGURE 7. Switching time waveforms types 01 and 02.

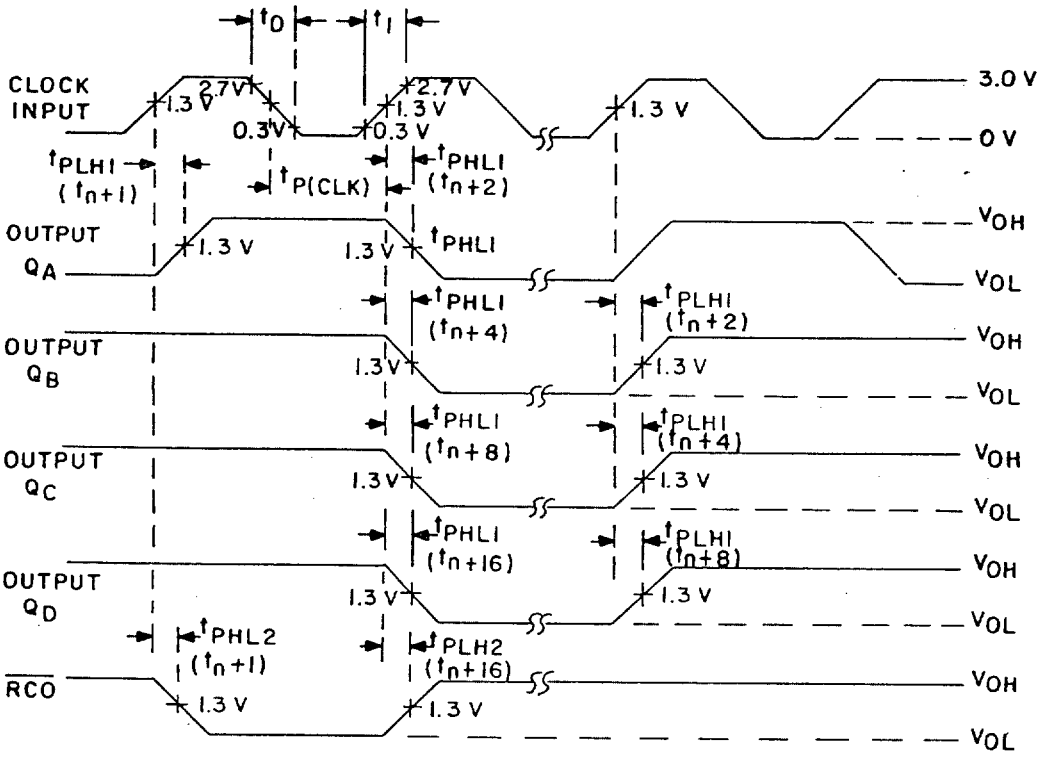
Source: https://assist.dla.mil -- Downloaded: 202610-21T14:34Z
Check the source to verify that this is the current version before use.



NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 20 ns; PRR $\leq$ 1 MHz.
2. For f_{MAX}, the clock input pulse characteristics are as follows: $t_1 = t_0 = 3 \pm 1.5$ ns; for 25°C, t_p (clock) = 12.5 ns; PRR = 30 MHz; for -55/125°C, t_p (clock) = 20 ns; PRR = 25 MHz.
3. Enable input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns;
4. Up/Down input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns;
5. Inputs not under test are at ground.
6. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
7. R_L = 499Ω±1%.
8. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 8. Switching time test circuit and waveforms for device type 03.

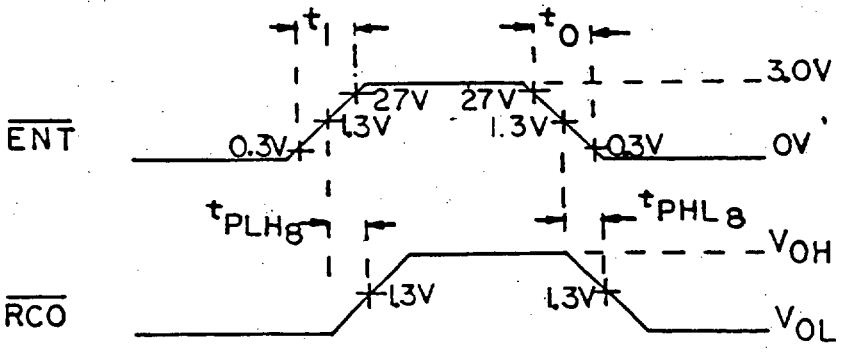
Source: https://assist.dla.mil -- Downloaded: 202610-21T14:34Z
Check the source to verify that this is the currentversion before use.



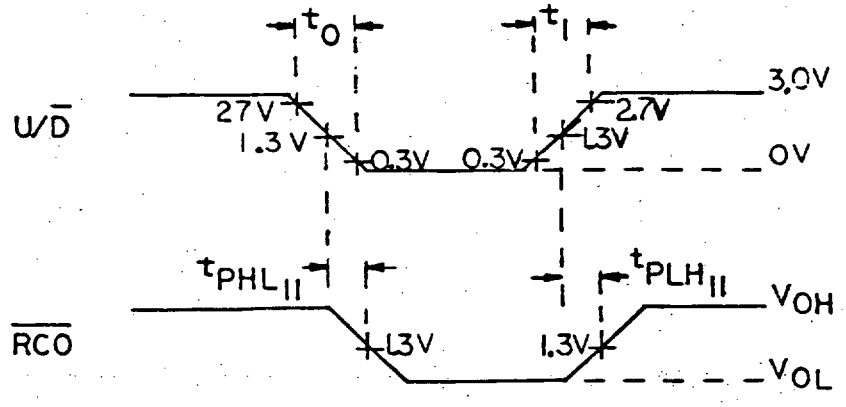
CLOCK TO OUTPUT

FIGURE 8. Switching time test circuit and waveforms for device type 03 - Continued.

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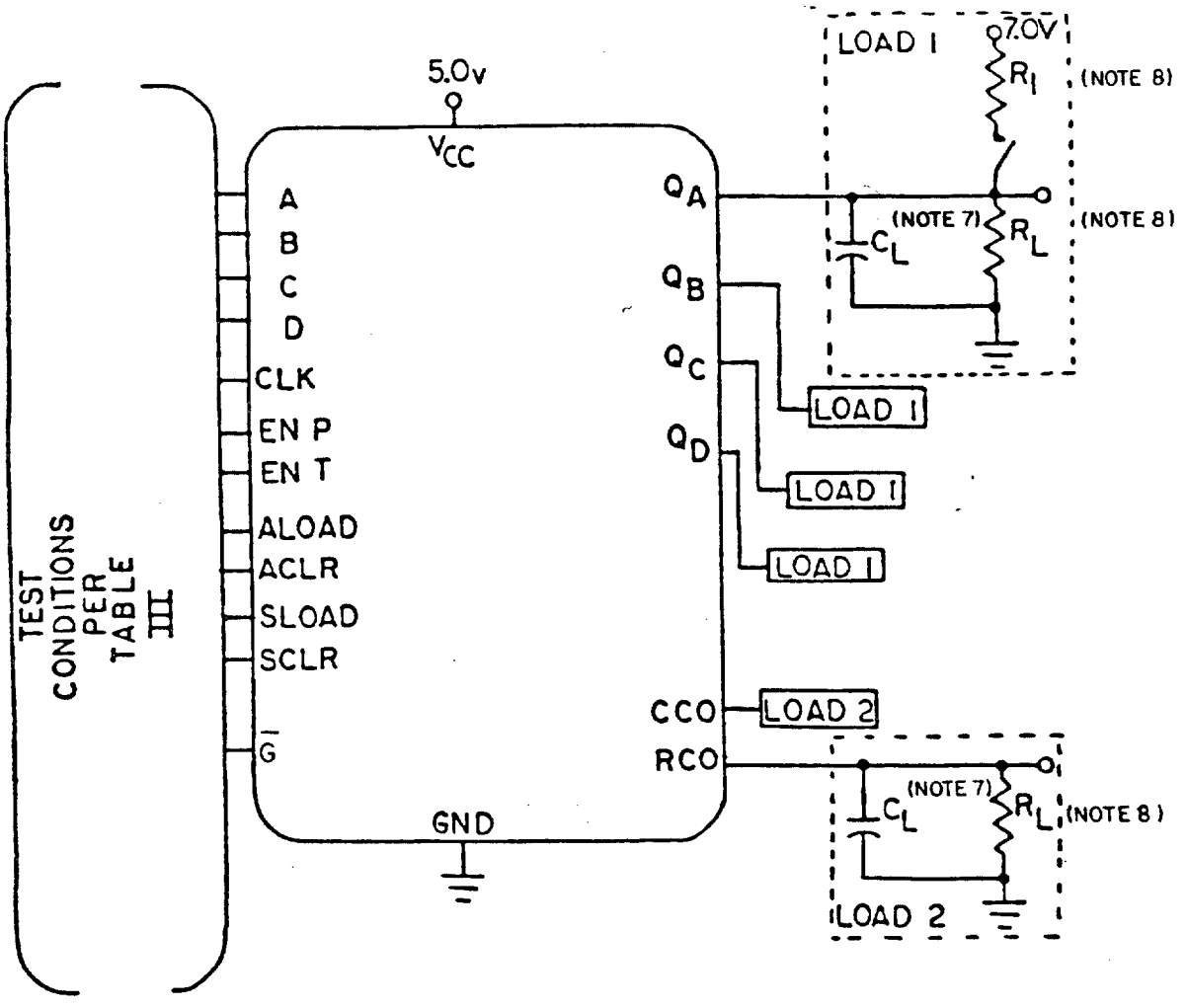
ENABLE TO RCO



UP/$\overline{\text{DOWN}}$ TO RCO

FIGURE 8. Switching time test circuit and waveforms for device type 03 - Continued.

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Check the source to verify that this is the currentversion before use.



NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 20 ns; PRR $\leq$ 1 MHz.
2. For f_{MAX}, the clock input pulse characteristics are as follows: $t_1 = t_0 = 3 \pm 1.5$ ns; for 25°C, t_p (clock) = 16.5 ns; PRR = 30 MHz; for -55/125°C, t_p (clock) = 20 ns; PRR = 25 MHz.
3. Asynchronous clear/load input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (ACLR/ALOAD) = 20 ns.
4. Enable input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_{setup} = 25 ns; t_{hold} = 0 ns; t_p (enable) = 25 ns.
5. Data input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_{setup} = 25 ns; t_{hold} = 0 ns; t_p (data) = 25 ns.
6. Output control input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p = 200 ns.
7. C_L = 50 pF ±10%, including scope, probe, wiring, and stray capacitance, without package in test fixture.
8. $R_1 = R_L = 499\Omega \pm 1\%$.
9. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 9. Switching time test circuit and waveforms for device type 04.

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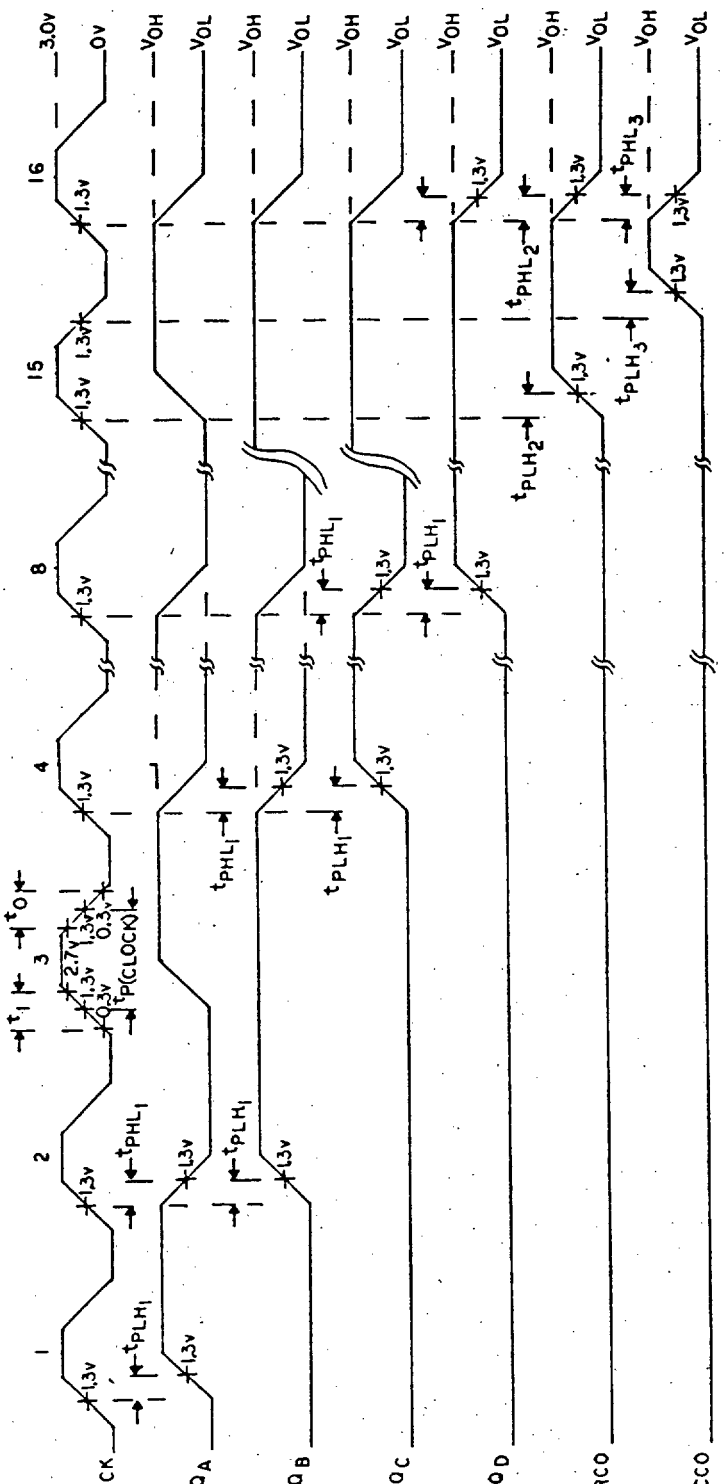
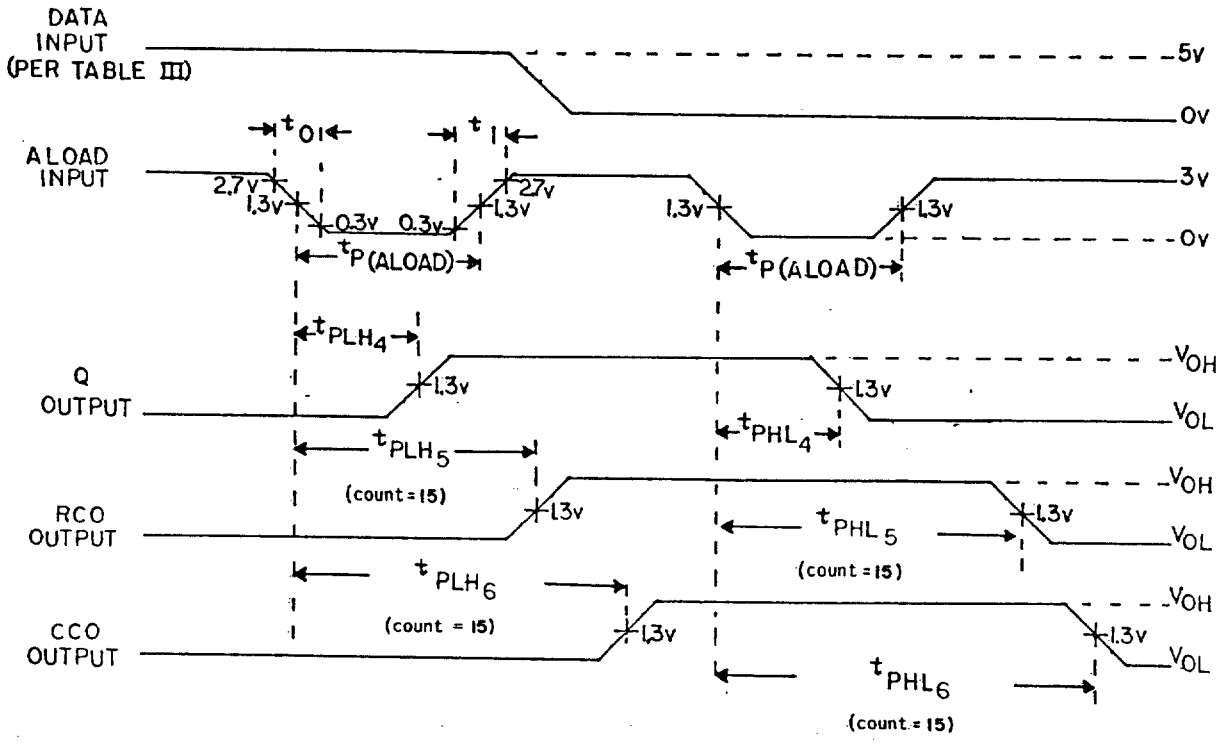


FIGURE 9. Switching time test circuit and waveforms device type 04 - Continued.

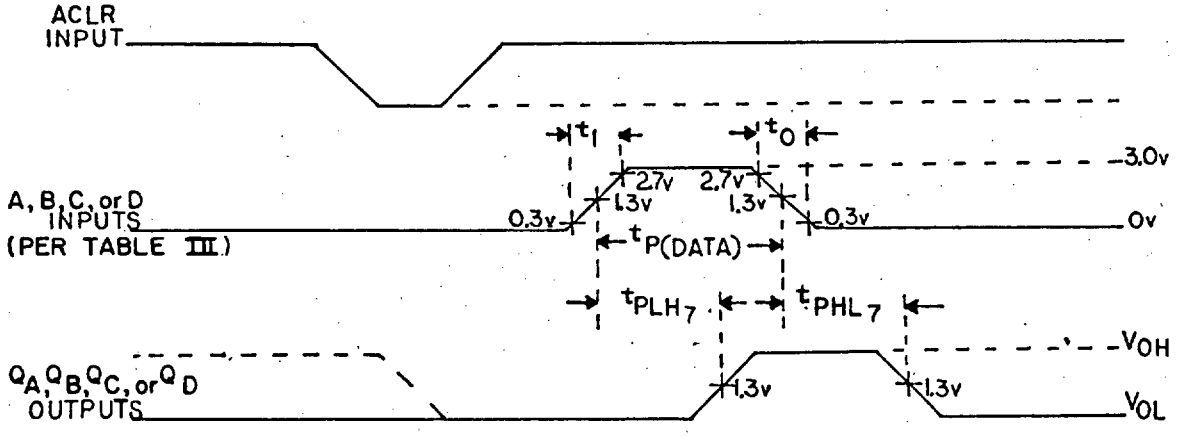
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ASYNCHRONOUS LOAD

FIGURE 9. Switching time test circuit and waveforms device type 04 - Continued.

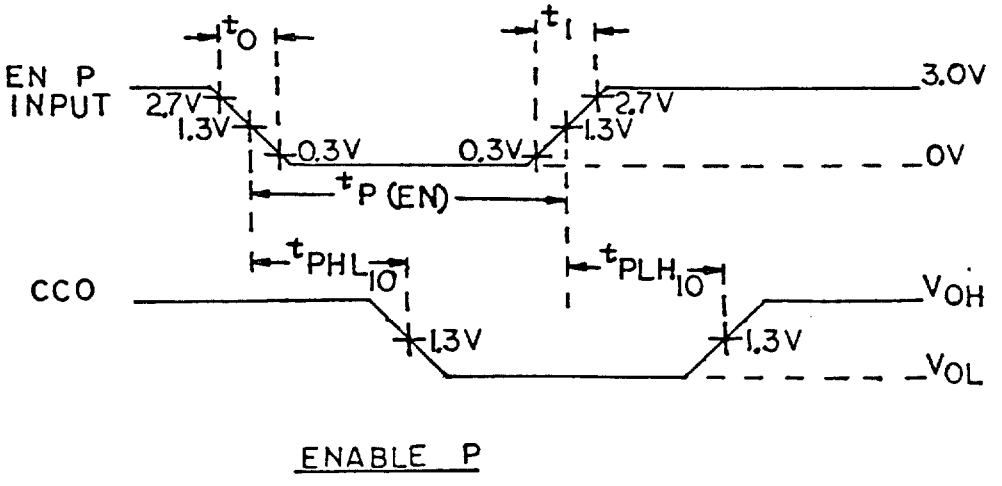
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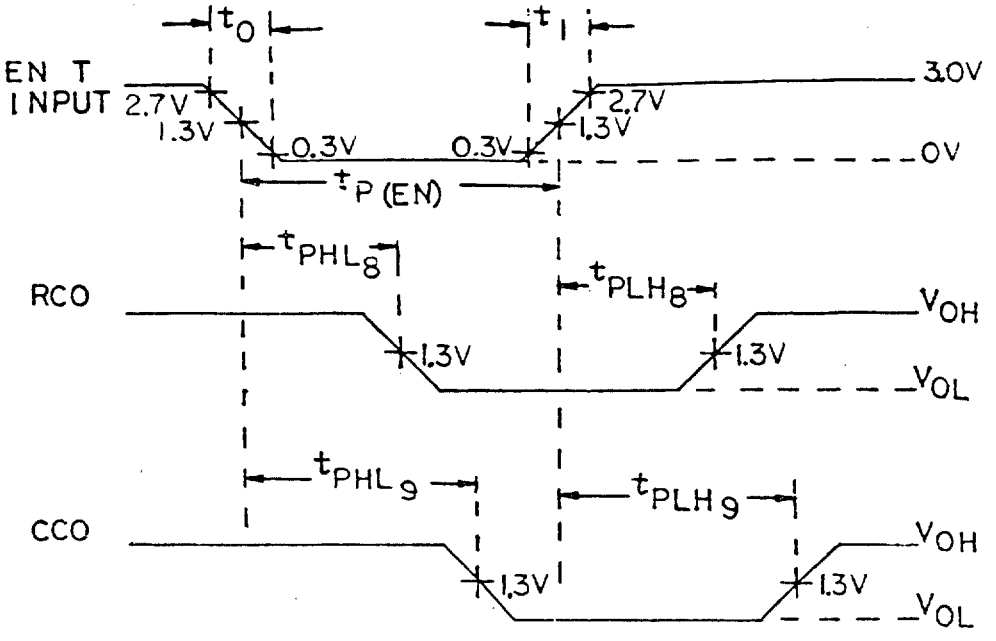
DATA TO Q OUTPUTS

FIGURE 9. Switching time test circuit and waveforms device type 04 - Continued.

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ENABLE P

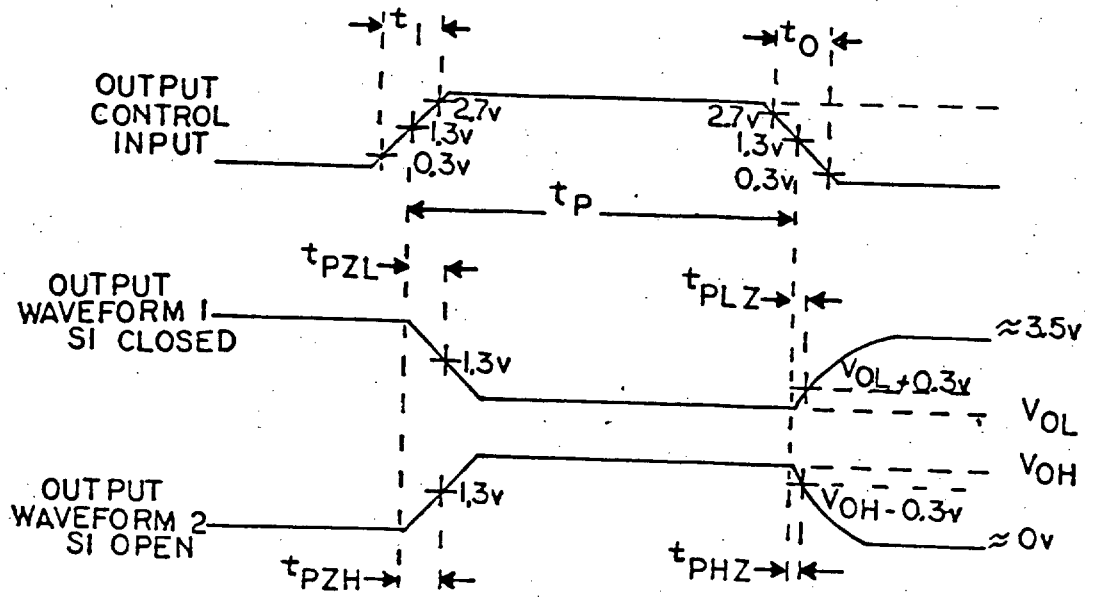


ENABLE T

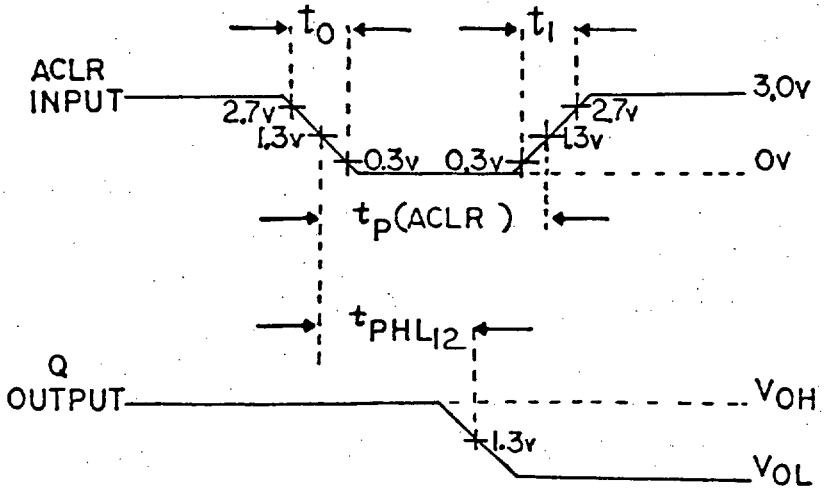
FIGURE 9. Switching time test circuit and waveforms device type 04 - Continued.

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Device type 04



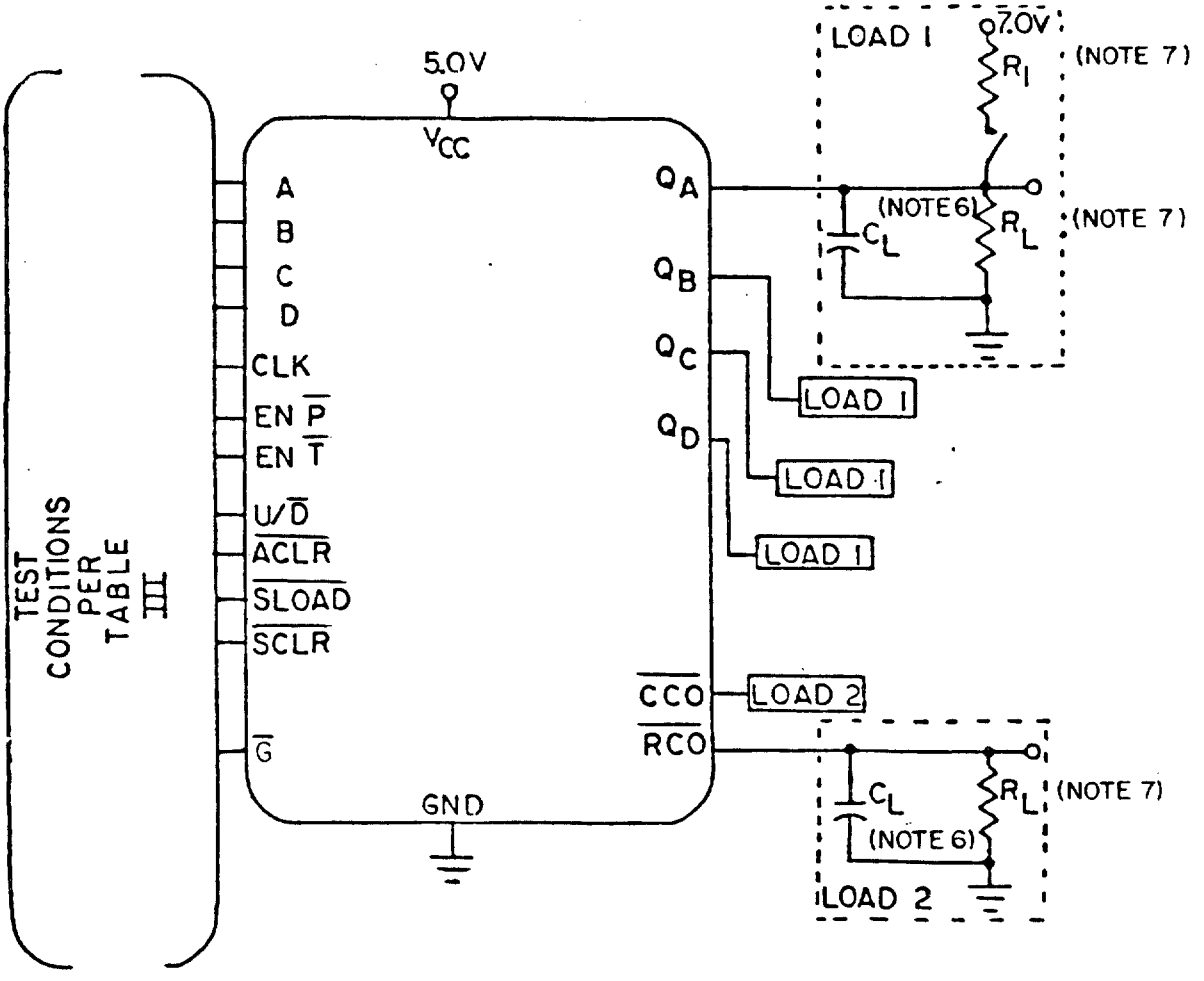
TRI-STATE SWITCHING



ASYNCHRONOUS CLEAR

FIGURE 9. Switching time test circuit and waveforms device type 04 - Continued.

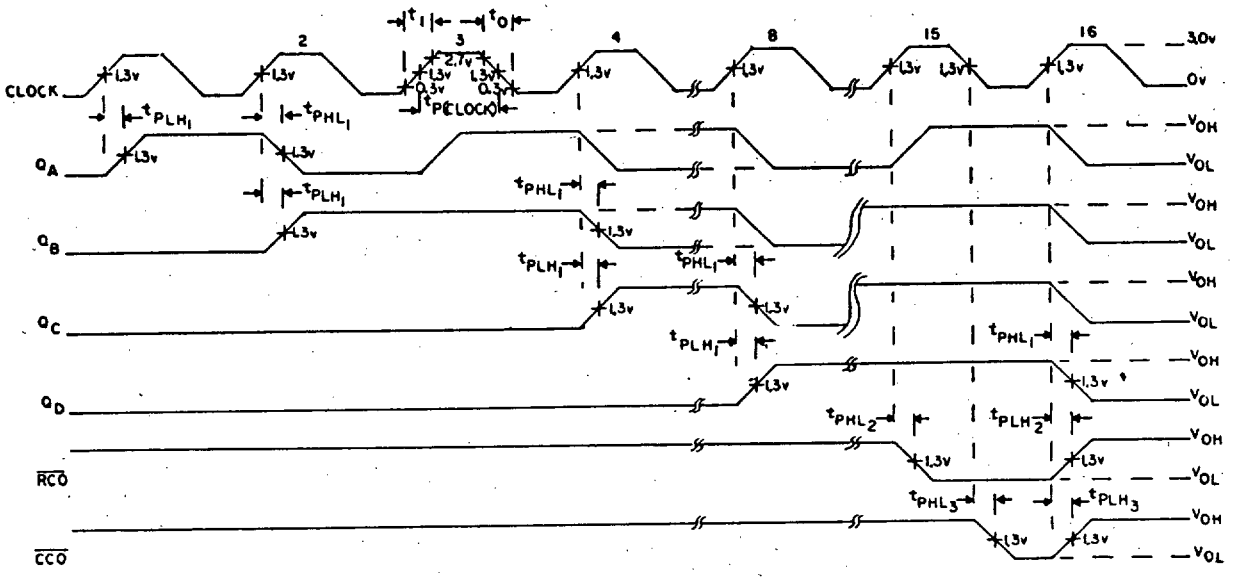
Source: https://assist.dla.mil -- Downloaded: 202610-21T14:34Z
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NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 20 ns; PRR $\leq$ 1 MHz.
2. For f_{MAX}, the clock input pulse characteristics are as follows: $t_1 = t_0 = 3 \pm 1.5$ ns; for 25°C, t_p (clock) = 16.5 ns; PRR = 30 MHz; for -55/125°C, t_p (clock) = 20 ns; PRR = 25 MHz.
3. Up/Down input pulse characteristics: $t_1 = 6 \pm 1.5$ ns; t_{setup} = 35 ns; t_{hold} = 0 ns; t_p (up/down) = 35 ns.
4. Enable input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; pulse high, t_{setup} = 35 ns; t_{hold} = 0 ns; t_p (enable) = 35 ns; pulse low, t_{setup} = 25 ns; t_p enable = 25 ns.
5. Asynchronous Clear input pulse characteristics; $t_1 = t_0 = 6 \pm 1.5$ ns; t_p ($\overline{ACLR}$) = 20 ns.
6. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
7. $R_1 = R_L = 499\,\Omega \pm 1\%$.
8. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 10. Switching time test circuit and waveforms for device type 05.

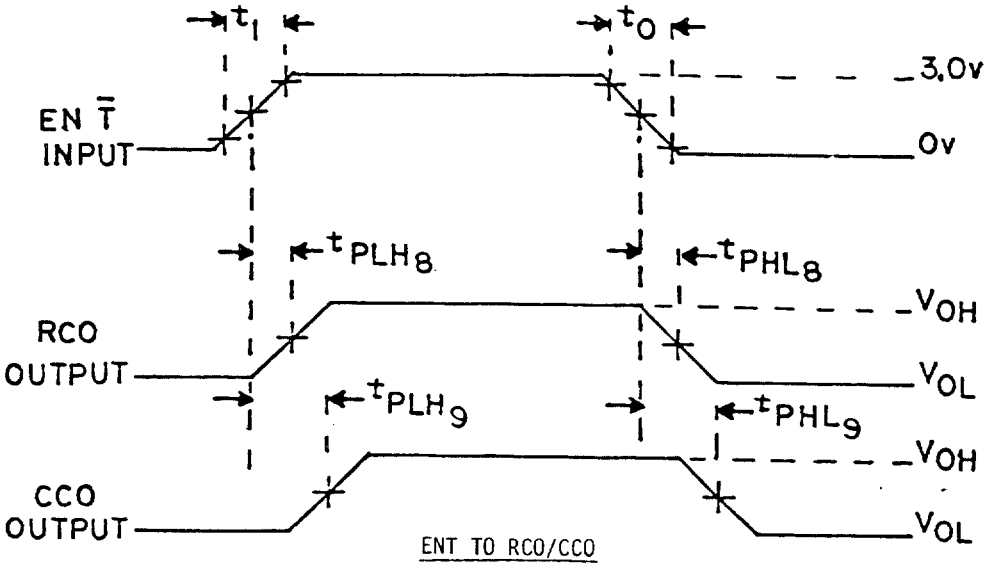
Source: https://assist.dla.mil -- Downloaded: 202610-21T14:34Z
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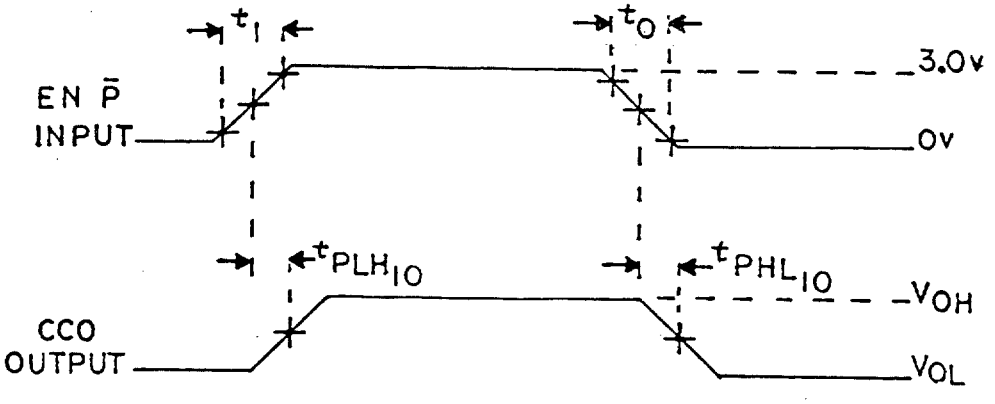
COUNT UP

FIGURE 10. Switching time test circuit and waveforms for device type 05 - Continued.

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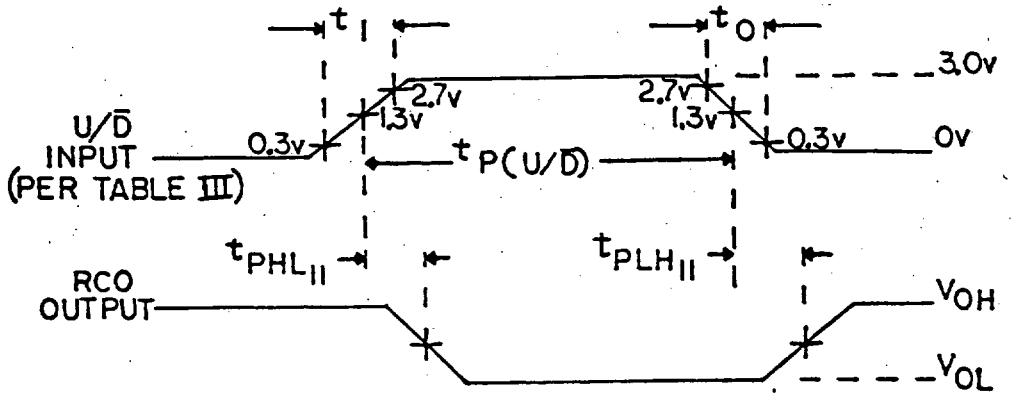
ENT TO RCO/CCO



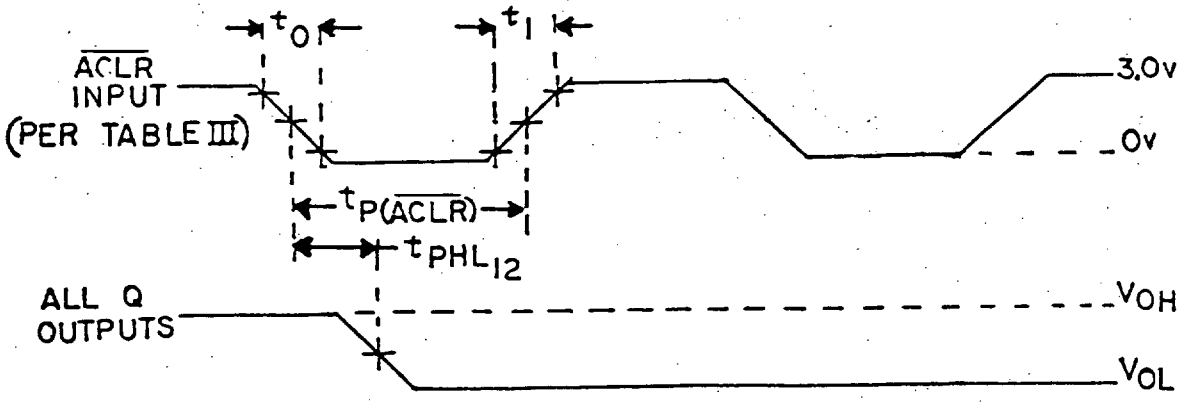
ENP TO CCO

FIGURE 10. Switching time test circuit and waveforms for device type 05 - Continued.

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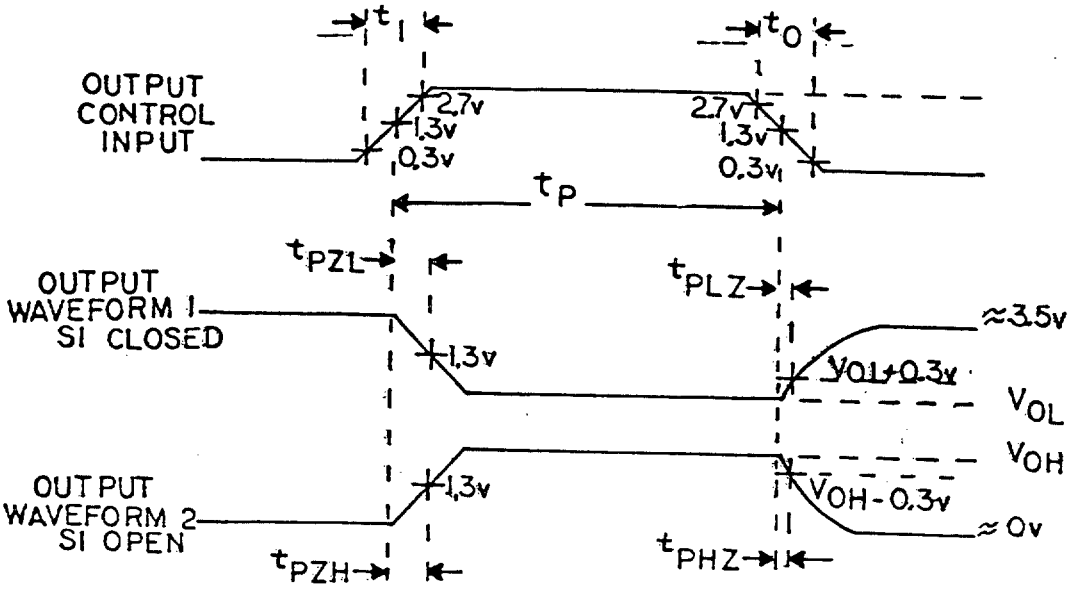
U/D̄ TO RCO̅



ACLR

FIGURE 10. Switching time test circuit and waveforms for device type 05 - Continued.

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TRI-STATE SWITCHING

FIGURE 10. Switching time test circuit and waveforms for device type 05 - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 $\overline{CLR}$	2 3 CLK	3 4 A	4 5 B	5 7 C	6 8 D	7 9 ENP	8 10 GND	9 12 $\overline{LOAD}$	10 13 ENT	11 14 QD	12 15 QC	13 17 QB	14 18 QA	15 19 RCO	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	2/														4.5 V	QD	2.5		V
			2	"	"			2.0 V							-400 µA				"	QC	"		"
			3	"	"		2.0 V									-400 µA			"	QB	"		"
			4	"	"	2.0 V											-400 µA		"	QA	"		"
			5	"	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	0.8 V	2.0 V	-400 µA				-400 µA	"	RCO	"		"
	V_{OL1}	3007	6	"	"						"	"	"						"	QD		0.4	"
			7	"	"			0.8 V			"	"	"	4 mA					"	QC		"	"
			8	"	"		0.8 V				"	"	"		4 mA				"	QB		"	"
			9	"	"	0.8 V					"	"	"			4 mA			"	QA		"	"
			10	"	"	0.8 V	0.8 V	0.8 V	0.8 V	"	"	"	"				4 mA		"	RCO		"	"
	V_{IC}		11	-18 mA							"								"	$\overline{CLR}$		-1.5	"
			12		-18 mA						"								"	CLK		"	"
			13			-18 mA					"								"	A		"	"
			14				-18 mA				"								"	B		"	"
			15					-18 mA			"								"	C		"	"
			16						-18 mA		"								"	D		"	"
			17							-18 mA	"								"	ENP		"	"
			18								"	-18 mA							"	$\overline{LOAD}$		"	"
			19								"		-18 mA						"	ENT		"	"
	I_{IL}	3009	20	0.4 V							"								5.5 V	$\overline{CLR}$	3/	3/	µA
			21		0.4 V						"								"	CLK	"	"	"
			22			0.4 V					"								"	A	"	"	"
			23				0.4 V				"								"	B	"	"	"
			24					0.4 V			"								"	C	"	"	"
			25						0.4 V		"								"	D	"	"	"
			26							0.4 V	"								"	ENP	"	"	"
			27								"	0.4 V							"	$\overline{LOAD}$	"	"	"
			28								"		0.4 V						"	ENT	"	"	"
	I_{IH1}	3010	29		2.7 V						"								"	CLK		40	"
			30								"	2.7 V							"	$\overline{LOAD}$		"	"
			31								"		2.7 V						"	ENT		"	"
			32	2.7 V							"								"	$\overline{CLR}$		20	"
			33			2.7 V					"								"	A		"	"
			34				2.7 V				"								"	B		"	"
			35					2.7 V			"								"	C		"	"
			36						2.7 V		"								"	D		"	"
			37							2.7 V	"								"	ENP		"	"
	I_{IH2}		38		7.0 V						"								"	CLK		200	"
			39								"	7.0 V							"	$\overline{LOAD}$		"	"
			40								"		7.0 V						"	ENT		"	"
			41	7.0 V							"								"	$\overline{CLR}$		100	"
			42			7.0 V					"								"	A		"	"
			43				7.0 V				"								"	B		"	"
			44					7.0 V			"								"	C		"	"
			45						7.0 V		"								"	D		"	"
			46							7.0 V	"								"	ENP		"	"
	I_O 4/	3011	47	5.5 V	2/	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	GND	5.5 V	2.25 V					"	QD	-30	-112	mA
			48	"	"	"	"	"	"	"	"	"	"		2.25 V				"	QC	"	"	"
			49	"	"	"	"	"	"	"	"	"	"			2.25 V			"	QB	"	"	"
			50	"	"	"	"	"	"	"	"	"	"				2.25 V		"	QA	"	"	"
			51	"	"	"	"	"	"	"	"	"	"					2.25 V	"	RCO	5/	5/	"
	I_{CC}	3005	52	"	"	"	"	"	"	"	"	"	"						"	V_{CC}		25	"
			53	"	"	GND	GND	GND	GND	"	"	"	"						"	V_{CC}		25	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				$\overline{CLR}$	CLK	A	B	C	D	ENP	GND	$\overline{LOAD}$	ENT	QD	QC	QB	QA	RCO	V_{CC}				
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C +25°C	Truth table test 6/	3014 Fig. 3	54	B 7/	B	B	B	B	B	B	B	B	B	L	L	L	L	L	5.0 V	8/			
		"	55	A 7/	B	B	B	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	56	A	A	B	B	B	B	A	B	A	A	L	L	L	H	L	"	"			
		"	57	A	B	B	B	B	B	A	B	A	A	L	L	L	H	L	"	"			
		"	58	A	A	B	B	B	B	A	B	A	A	L	L	H	L	L	"	"			
		"	59	A	B	B	B	B	B	A	B	A	A	L	L	H	L	L	"	"			
		"	60	A	A	B	B	B	B	A	B	A	A	L	L	H	H	L	"	"			
		"	61	A	B	B	B	B	B	A	B	A	A	L	L	H	H	L	"	"			
		"	62	A	A	B	B	B	B	A	B	A	A	L	H	L	L	L	"	"			
		"	63	A	B	B	B	B	B	A	B	A	A	L	H	L	L	L	"	"			
		"	64	A	A	B	B	B	B	A	B	A	A	L	H	L	H	L	"	"			
		"	65	A	B	B	B	B	B	A	B	A	A	L	H	L	H	L	"	"			
		"	66	A	A	B	B	B	B	A	B	A	A	L	H	H	L	L	"	"			
		"	67	A	B	B	B	B	B	A	B	A	A	L	H	H	L	L	"	"			
		"	68	A	A	B	B	B	B	A	B	A	A	L	H	H	H	L	"	"			
		"	69	A	B	B	B	B	B	A	B	A	A	L	H	H	H	L	"	"			
		"	70	A	A	B	B	B	B	A	B	A	A	H	L	L	L	L	"	"			
		"	71	A	A	B	B	B	B	B	B	A	A	H	L	L	L	L	"	"			
		"	72	A	B	B	B	B	B	B	B	A	A	H	L	L	L	L	"	"			
		"	73	A	A	B	B	B	B	B	B	A	A	H	L	L	L	L	"	"			
		"	74	A	B	B	B	B	B	A	B	A	A	H	L	L	L	L	"	"			
		"	75	A	A	B	B	B	B	A	B	A	A	H	L	L	L	H	"	"			
		"	76	A	B	B	B	B	B	A	B	A	A	H	L	L	H	L	"	"			
		"	77	A	A	B	B	B	B	A	B	A	A	H	L	H	L	L	"	"			
		"	78	A	B	B	B	B	B	A	B	A	A	H	L	H	L	L	"	"			
		"	79	A	A	B	B	B	B	A	B	A	A	H	L	H	H	L	"	"			
		"	80	A	B	B	B	B	B	A	B	A	A	H	L	H	H	L	"	"			
		"	81	A	A	B	B	B	B	A	B	A	A	H	H	L	L	L	"	"			
		"	82	A	B	B	B	B	B	A	B	A	A	H	H	L	L	L	"	"			
		"	83	A	A	B	B	B	B	A	B	A	A	H	H	L	H	L	"	"			
		"	84	A	B	B	B	B	B	A	B	A	A	H	H	L	H	L	"	"			
		"	85	A	A	B	B	B	B	A	B	A	A	H	H	H	L	L	"	"			
		"	86	A	B	B	B	B	B	A	B	A	A	H	H	H	L	L	"	"			
		"	87	A	A	B	B	B	B	A	B	A	A	H	H	H	H	H	"	"			
		"	88	A	B	B	B	B	B	A	B	A	A	H	H	H	H	H	"	"			
		"	89	A	B	B	B	B	B	A	B	A	B	H	H	H	H	L	"	"			
		"	90	A	A	B	B	B	B	A	B	A	B	H	H	H	H	L	"	"			
		"	91	A	A	B	B	B	B	A	B	A	A	H	H	H	H	H	"	"			
		"	92	A	B	B	B	B	B	A	B	A	A	H	H	H	H	H	"	"			
		"	93	A	A	B	B	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	94	A	B	B	B	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	95	A	B	B	B	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	96	A	B	A	B	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	97	A	B	A	A	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	98	A	B	A	A	A	B	A	B	A	A	L	L	L	L	L	"	"			
		"	99	A	B	A	A	A	B	A	B	B	A	L	L	L	L	L	"	"			
		"	100	A	A	A	A	A	B	A	B	B	A	L	H	H	L	L	"	"			
		"	101	A	A	A	A	A	B	A	B	A	A	L	H	H	H	L	"	"			
		"	102	A	B	A	A	A	B	A	B	A	A	L	H	H	H	L	"	"			
		"	103	A	A	A	A	A	B	A	B	A	A	H	L	L	L	L	"	"			
		"	104	A	B	A	A	A	B	A	B	A	A	H	L	L	L	L	"	"			
		"	105	B	B	A	A	A	A	A	B	A	A	L	L	L	L	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	$\overline{CLR}$	CLK	A	B	C	D	ENP	GND	$\overline{LOAD}$	ENT	QD	QC	QB	QA	RCO	V_{CC}				
9 T_C = +25°C	f_{MAX} 9/	3003 Fig. 6	106	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	CLK to QA	40		MHz
	t_{PLH1}		107	"	"					"	"	"	"						"	CLK to QA	4	15	ns
			108	"	"					"	"	"	"						"	CLK to QB	"	"	"
			109	"	"					"	"	"	"						"	CLK to QC	"	"	"
			110	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CLK to QD	"	"	"
	t_{PHL1}		111	"	"					"	"	"	"						"	CLK to QA	6	20	"
			112	"	"					"	"	"	"						"	CLK to QB	"	"	"
			113	"	"					"	"	"	"						"	CLK to QC	"	"	"
			114	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CLK to QD	"	"	"
	t_{PLH2}		115	"	"					"	"	"	"					OUT	"	CLK to RCO	5	"	"
	t_{PHL2}		116	"	"					"	"	"	"					"	"	CLK to RCO	5	"	"
	t_{PLH8}		117	"						"	"	"	IN					"	"	ENT to RCO	3	13	"
	t_{PHL8}		118	"						"	"	"	IN					"	"	ENT to RCO	3	13	"
	t_{PHL12}		119	IN	2/	4.5 V	4.5 V	4.5 V	4.5 V		"	GND							"	$\overline{CLR}$ to QA	8	24	"
			120	"	"						"	"							"	$\overline{CLR}$ to QB	"	"	"
			121	"	"						"	"							"	$\overline{CLR}$ to QC	"	"	"
			122	"	"						"	"		OUT	OUT	OUT	OUT		"	$\overline{CLR}$ to QD	"	"	"
	t_{PHL13}		123	"	"	4.5 V	4.5 V	4.5 V	4.5 V		"	"						OUT	"	$\overline{CLR}$ to RCO	11	23	"
10 T_C = +125°C	f_{MAX} 9/		Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																		35		MHz
	t_{PLH1}																				4	18	ns
	t_{PHL1}																				6	25	"
	t_{PLH2}																				5	"	"
	t_{PHL2}																				5	"	"
	t_{PLH8}																				3	16	"
	t_{PHL8}																				3	16	"
	t_{PHL12}																				8	27	"
	t_{PHL13}																				11	28	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / Test no.	1 / 2	2 / 3	3 / 4	4 / 5	5 / 7	6 / 8	7 / 9	8 / 10	9 / 12	10 / 13	11 / 14	12 / 15	13 / 17	14 / 18	15 / 19	16 / 20	Measured terminal	Limits Min	Limits Max	Unit
				$\overline{CLR}$	CLK	A	B	C	D	ENP	GND	$\overline{LOAD}$	ENT	QD	QC	QB	QA	RCO	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	2/			2.0 V	2.0 V	2.0 V	GND	0.8 V	2.0 V	-400 µA					4.5 V	QD	2.5		V
			2	"	"		2.0 V			"	"	"	"		-400 µA				"	QC	"		"
			3	"	"			2.0 V		"	"	"	"			-400 µA			"	QB	"		"
			4	"	"	2.0 V				"	"	"	"				-400 µA		"	QA	"		"
			5	"	"	2.0 V	2.0 V	2.0 V	2.0 V	"	"	"	"					-400 µA	"	RCO	"		"
	V_{OL1}	3007	6	"	"				0.8 V	"	"	"	"	4 mA					"	QD		0.4	"
			7	"	"			0.8 V		"	"	"	"		4 mA				"	QC		"	"
			8	"	"		0.8 V			"	"	"	"			4 mA			"	QB		"	"
			9	"	"	0.8 V				"	"	"	"				4 mA		"	QA		"	"
			10	"	"	0.8 V	0.8 V	0.8 V	0.8 V	"	"	"	"					4 mA	"	RCO		"	"
	V_{IC}		11	-18 mA							"								"	$\overline{CLR}$		-1.5	"
			12		-18 mA						"								"	CLK		"	"
			13			-18 mA					"								"	A		"	"
			14				-18 mA				"								"	B		"	"
			15					-18 mA			"								"	C		"	"
			16						-18 mA		"								"	D		"	"
			17							-18 mA	"								"	ENP		"	"
			18								"	-18 mA							"	$\overline{LOAD}$		"	"
			19								"		-18 mA						"	ENT		"	"
	I_{IL}	3009	20	0.4 V							"								5.5 V	$\overline{CLR}$	3/	3/	µA
			21		0.4 V						"								"	CLK	"	"	"
			22			0.4 V					"								"	A	"	"	"
			23				0.4 V				"								"	B	"	"	"
			24					0.4 V			"								"	C	"	"	"
			25						0.4 V		"								"	D	"	"	"
			26							0.4 V	"								"	ENP	"	"	"
			27								"	0.4 V							"	$\overline{LOAD}$	"	"	"
			28								"		0.4 V						"	ENT	"	"	"
	I_{IH1}	3010	29		2.7 V						"	2.7 V							"	CLK		40	"
			30								"		2.7 V						"	$\overline{LOAD}$		"	"
			31								"								"	ENT		"	"
			32	2.7 V							"								"	$\overline{CLR}$		20	"
			33			2.7 V					"								"	A		"	"
			34				2.7 V				"								"	B		"	"
			35					2.7 V			"								"	C		"	"
			36						2.7 V		"								"	D		"	"
			37							2.7 V	"								"	ENP		"	"
	I_{IH2}		38		7.0 V						"	7.0 V							"	CLK		200	"
			39								"		7.0 V						"	$\overline{LOAD}$		"	"
			40								"								"	ENT		"	"
			41	7.0 V							"								"	$\overline{CLR}$		100	"
			42			7.0 V					"								"	A		"	"
			43				7.0 V				"								"	B		"	"
			44					7.0 V			"								"	C		"	"
			45						7.0 V		"								"	D		"	"
			46							7.0 V	"								"	ENP		"	"
	I_O 4/	3011	47	5.5 V	2/	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	GND	5.5 V	2.25 V					"	QD	-30	-112	mA
			48	"	"	"	"	"	"	"	"	"	"		2.25 V				"	QC	"	"	"
			49	"	"	"	"	"	"	"	"	"	"			2.25 V			"	QB	"	"	"
			50	"	"	"	"	"	"	"	"	"	"				2.25 V		"	QA	"	"	"
			51	"	"	"	"	"	"	"	"	"	"					2.25 V	"	RCO	-15	-70	"
	I_{CC}	3005	52	"	"	"	"	"	"	"	"	"	"						"	V_{CC}		25	"
			53	"	"	GND	GND	GND	GND	"	"	"	"						"	V_{CC}		25	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Cases E, F Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			test no.	$\overline{CLR}$	CLK	A	B	C	D	ENP	GND	$\overline{LOAD}$	ENT	QD	QC	QB	QA	RCO	V_{CC}				
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C +25°C	Truth table test 6/	3014 Fig. 3	54	B	B	B	B	B	B	B	B	B	B	L	L	L	L	L	5.0 V	8/			
		"	55	B	A	B	B	B	B	B	B	B	B	L	L	L	L	L	"	"			
		"	56	B	B	B	B	B	B	B	B	B	B	L	L	L	L	L	"	"			
		"	57	A	B	B	B	B	B	B	B	B	B	L	L	L	L	L	"	"			
		"	58	A	B	B	B	B	B	A	B	B	B	L	L	L	L	L	"	"			
		"	59	A	B	B	B	B	B	A	B	A	B	L	L	L	L	L	"	"			
		"	60	A	B	B	B	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	61	A	A	B	B	B	B	A	B	A	A	L	L	L	H	L	"	"			
		"	62	A	B	B	B	B	B	A	B	A	A	L	L	L	H	L	"	"			
		"	63	A	A	B	B	B	B	A	B	A	A	L	L	H	L	L	"	"			
		"	64	A	B	B	B	B	B	A	B	A	A	L	L	H	L	L	"	"			
		"	65	A	A	B	B	B	B	A	B	A	A	L	L	H	H	L	"	"			
		"	66	A	B	B	B	B	B	A	B	A	A	L	L	H	H	L	"	"			
		"	67	A	A	B	B	B	B	A	B	A	A	L	H	L	L	L	"	"			
		"	68	A	B	B	B	B	B	A	B	A	A	L	H	L	L	L	"	"			
		"	69	A	A	B	B	B	B	A	B	A	A	L	H	L	H	L	"	"			
		"	70	A	B	B	B	B	B	A	B	A	A	L	H	L	H	L	"	"			
		"	71	A	A	B	B	B	B	A	B	A	A	L	H	H	L	L	"	"			
		"	72	A	B	B	B	B	B	A	B	A	A	L	H	H	L	L	"	"			
		"	73	A	A	B	B	B	B	A	B	A	A	L	H	H	H	L	"	"			
		"	74	A	B	B	B	B	B	A	B	A	A	L	H	H	H	L	"	"			
		"	75	A	A	B	B	B	B	A	B	A	A	H	L	L	L	L	"	"			
		"	76	A	B	B	B	B	B	A	B	A	A	H	L	L	L	L	"	"			
		"	77	A	B	B	B	B	B	9	B	A	A	H	L	L	L	L	"	"			
		"	78	A	A	B	B	B	B	B	B	A	A	H	L	L	L	L	"	"			
		"	79	A	B	B	B	B	B	A	B	A	A	H	L	L	L	L	"	"			
		"	80	A	A	B	B	B	B	A	B	A	A	H	L	L	H	L	"	"			
		"	81	A	B	B	B	B	B	A	B	A	A	H	L	L	H	L	"	"			
		"	82	A	A	B	B	B	B	A	B	A	A	H	L	H	L	L	"	"			
		"	83	A	B	B	B	B	B	A	B	A	A	H	L	H	L	L	"	"			
		"	84	A	A	B	B	B	B	A	B	A	A	H	L	H	H	L	"	"			
		"	85	A	B	B	B	B	B	A	B	A	A	H	L	H	H	L	"	"			
		"	86	A	A	B	B	B	B	A	B	A	A	H	H	L	L	L	"	"			
		"	87	A	B	B	B	B	B	A	B	A	A	H	H	L	L	L	"	"			
		"	88	A	A	B	B	B	B	A	B	A	A	H	H	L	H	L	"	"			
		"	89	A	B	B	B	B	B	A	B	A	A	H	H	L	H	L	"	"			
		"	90	A	A	B	B	B	B	A	B	A	A	H	H	H	L	L	"	"			
		"	91	A	B	B	B	B	B	A	B	A	A	H	H	H	L	L	"	"			
		"	92	A	A	B	B	B	B	A	B	A	A	H	H	H	H	H	"	"			
		"	93	A	B	B	B	B	B	A	B	A	A	H	H	H	H	H	"	"			
		"	94	A	B	B	B	B	B	A	B	A	B	H	H	H	H	H	"	"			
		"	95	A	A	B	B	B	B	A	B	A	B	H	H	H	H	H	"	"			
		"	96	A	B	B	B	B	B	A	B	A	B	H	H	H	H	H	"	"			
		"	97	A	B	B	B	B	B	A	B	A	A	H	H	H	H	H	"	"			
		"	98	A	A	B	B	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	99	A	B	B	B	B	B	A	B	A	A	L	L	L	L	L	"	"			
		"	100	A	B	B	B	A	B	A	B	A	A	L	L	L	L	L	"	"			
		"	101	A	B	B	B	A	A	A	B	A	A	L	L	L	L	L	"	"			
		"	102	A	B	B	B	A	A	A	B	A	A	L	L	L	L	L	"	"			
		"	103	A	B	B	B	A	A	A	B	B	A	L	L	L	L	L	"	"			
		"	104	A	A	B	B	A	A	A	B	B	A	H	H	H	L	L	"	"			
		"	105	A	A	B	B	A	A	A	B	A	A	H	H	H	L	L	"	"			
		"	106	A	B	B	B	A	A	A	B	A	A	H	H	H	L	L	"	"			
		"	107	A	A	B	B	A	A	A	B	A	A	H	H	H	H	H	"	"			
		"	108	A	B	B	B	A	A	A	B	A	A	H	H	H	H	H	"	"			
		"	109	A	B	B	A	A	A	A	B	A	A	H	H	H	H	H	"	"			
		"	110	B	B	A	A	A	A	A	B	A	A	H	H	H	H	H	"	"			
		"	111	B	A	A	A	A	A	A	B	A	A	L	L	L	L	L	"	"			
		"	112	B	B	A	A	A	A	A	B	A	A	L	L	L	L	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

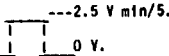
Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Test no.	$\overline{CLR}$	CLK	A	B	C	D	ENP	GND	$\overline{LOAD}$	ENT	QD	QC	QB	QA	RCO	V_{CC}		Min	Max	
9 T_C = +25°C	f_{MAX} 9/	3003 Fig. 6	113	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	CLK to QA	40		MHz
	t_{PLH1}		114	"	"					"	"	"	"				OUT		"	CLK to QA	4	15	ns
			115	"	"					"	"	"	"			OUT			"	CLK to QB	"	"	"
			116	"	"					"	"	"	"		OUT				"	CLK to QC	"	"	"
			117	"	"					"	"	"	"	OUT					"	CLK to QD	"	"	"
	t_{PHL1}		118	"	"					"	"	"	"				OUT		"	CLK to QA	6	20	"
			119	"	"					"	"	"	"			OUT			"	CLK to QB	"	"	"
			120	"	"					"	"	"	"		OUT				"	CLK to QC	"	"	"
			121	"	"					"	"	"	"	OUT					"	CLK to QD	"	"	"
	t_{PLH2}		122	"	"					"	"	"	"					OUT	"	CLK to RCO	5	"	"
	t_{PHL2}		123	"	"					"	"	"	"					"	"	CLK to RCO	5	"	"
	t_{PLH8}		124	"						"	"	"	IN					"	"	ENT to RCO	3	13	"
	t_{PHL8}		125	"						"	"	"	IN					"	"	ENT to RCO	3	13	"
10 T_C = +125°C	f_{MAX} 9/		Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																		25		MHz
	t_{PLH1}																				4	18	ns
	t_{PHL1}																				6	20	"
	t_{PLH2}																				8	30	"
	t_{PHL2}																				7	25	"
	t_{PLH8}																				5	20	"
	t_{PHL8}																				4	16	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 02.

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Device types 01 and 02

1/ Case 2 pins not referenced are N/C.

2/ Apply one clock pulse prior to test as follows:  ---2.5 V min/5.5 V max; 0 V.

3/ I_{IL} limits shall be as follows:

Test	Device type	Min/max limits in µA for circuit		
		A	B	C
I_{IL}	01	0/-200	0/-200	
	02	0/-200	0/-200	

4/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.

5/ I_O limits for RCO pin shall be as follows:

Test	Device type	Min/max limits in mA for circuit		
		A	B	C
I_O (RCO pin only)	01	-30/-112	-30/-112	
	02	-30/-112	-30/-112	

6/ Only a summary of attributes data is required.

7/ A = 3.0 V minimum; B = 0.0 V or GND; X = don't care.

8/ H >1.5 V; L <1.5 V.

9/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				U/D	CLK	A	B	C	D	ENP	GND	LOAD	ENT	QD	QC	QB	QA	RCO	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	2/	2.0 V				0.8 V	GND	0.8 V	0.8 V				-400 µA		4.5 V	QA	2.5		V
			2	"	"	"	2.0 V			"	"	"	"			-400 µA	"		"	QB	"		"
			3	"	"	"	"	2.0 V		"	"	"	"		-400 µA	"			"	QC	"		"
			4	"	"	"	"	"	2.0 V	"	"	"	"	-400 µA					"	QD	"		"
			5	"	"	0.8 V	0.8 V	0.8 V	0.8 V	"	"	"	"					-400 µA	"	RCO	"		"
	V_{OL1}	3007	6	"	"	0.8 V				"	"	"	"				4 mA		"	QA		0.4	"
			7	"	"	"	0.8 V			"	"	"	"			4 mA	"		"	QB		"	"
			8	"	"	"	"	0.8 V		"	"	"	"		4 mA	"	"		"	QC		"	"
			9	"	"	"	"	"	0.8 V	"	"	"	"	4 mA		"	"		"	QD		"	"
			10	"	"	2.0 V	2.0 V	2.0 V	2.0 V	"	"	"	"					4 mA	"	RCO		"	"
	V_{IC}		11	-18 mA	-18 mA						"								"	U/D		-1.5	"
			12			-18 mA					"								"	CLK		"	"
			13				-18 mA				"								"	A		"	"
			14					-18 mA			"								"	B		"	"
			15						-18 mA		"								"	C		"	"
			16							-18 mA	"								"	D		"	"
			17								"	-18 mA							"	ENP		"	"
			18								"		-18 mA						"	LOAD		"	"
			19								"								"	ENT		"	"
	I_{IL}	3009	20	0.4 V							"								5.5 V	U/D	3/	3/	µA
			21		0.4 V						"								"	CLK	"	"	"
			22			0.4 V					"								"	A	"	"	"
			23				0.4 V				"								"	B	"	"	"
			24					0.4 V			"								"	C	"	"	"
			25						0.4 V		"								"	D	"	"	"
			26							0.4 V	"								"	ENP	"	"	"
			27								"	0.4 V							"	LOAD	"	"	"
			28								"		0.4 V						"	ENT	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	U/D		20	"
			30		2.7 V						"								"	CLK		"	"
			31			2.7 V					"								"	A		"	"
			32				2.7 V				"								"	B		"	"
			33					2.7 V			"								"	C		"	"
			34						2.7 V		"								"	D		"	"
			35							2.7 V	"								"	ENP		"	"
			36								"	2.7 V							"	LOAD		"	"
			37								"		2.7 V						"	ENT		"	"
	I_{IH2}		38	7.0 V							"								"	U/D		100	"
			39		7.0 V						"								"	CLK		"	"
			40			7.0 V					"								"	A		"	"
			41				7.0 V				"								"	B		"	"
			42					7.0 V			"								"	C		"	"
			43						7.0 V		"								"	D		"	"
			44							7.0 V	"								"	ENP		"	"
			45								"	7.0 V							"	LOAD		"	"
			46								"		7.0 V						"	ENT		"	"
	I_O 4/	3011	47	4.5 V	2/	4.5 V				GND	"	GND	GND				2.25 V		"	QA	-30	-112	mA
			48	"	"	"	4.5 V			"	"	"	"			2.25 V	"		"	QB	"	"	"
			49	"	"	"	"	4.5 V		"	"	"	"		2.25 V	"	"		"	QC	"	"	"
			50	"	"	"	"	"	4.5 V	"	"	"	"	2.25 V	"	"	"		"	QD	"	"	"
			51	"	"	GND	GND	GND	GND	"	"	"	"					2.25 V	"	RCO	"	"	"
	I_{CC}	3005	52	"	"	4.5 V	4.5 V	4.5 V	"	"	"	"	"						"	V_{CC}		25	"
			53	"	"	GND	GND	GND	"	"	"	"	"						"	V_{CC}		25	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 U/D	2 3 CLK	3 4 A	4 5 B	5 7 C	6 8 D	7 9 $\overline{ENP}$	8 10 GND	9 12 $\overline{LOAD}$	10 13 $\overline{ENT}$	11 14 QD	12 15 QC	13 17 QB	14 18 QA	15 19 $\overline{RCO}$	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C +25°C	Truth table test 5/	3014 Fig. 3	54	A 6/	B	B	B	B	B	B	GND	B	B	X	X	X	X	X	5.0 V	7/			
			55	A	A	B	B	B	B	B	"	B	B	L	L	L	L	H	"	"			
			56	A	B	B	B	B	B	B	"	A	B	L	L	L	L	H	"	"			
			57	A	A	B	B	B	B	B	"	A	B	L	L	L	H	H	"	"			
			58	A	B	B	B	B	B	B	"	A	B	L	L	L	H	H	"	"			
			59	A	A	B	B	B	B	B	"	A	B	L	L	H	L	H	"	"			
			60	A	B	B	B	B	B	B	"	A	B	L	L	H	L	H	"	"			
			61	A	A	B	B	B	B	B	"	A	B	L	L	H	H	H	"	"			
			62	A	B	B	B	B	B	B	"	A	B	L	L	H	H	H	"	"			
			63	A	A	B	B	B	B	B	"	A	B	L	H	L	L	H	"	"			
			64	A	B	B	B	B	B	B	"	A	B	L	H		L	H	"	"			
			65	A	A	B	B	B	B	B	"	A	B	L	H	L	H	H	"	"			
			66	A	B	B	B	B	B	B	"	A	B	L	H	L	H	H	"	"			
			67	A	A	B	B	B	B	B	"	A	B	L	H	H	L	H	"	"			
			68	A	B	B	B	B	B	B	"	A	B	L	H	H	L	H	"	"			
			69	A	A	B	B	B	B	B	"	A	B	L	H	H	H	H	"	"			
			70	A	B	B	B	B	B	B	"	A	B	L	H	H	H	H	"	"			
			71	A	A	B	B	B	B	B	"	A	B	H	L	L	L	H	"	"			
			72	A	B	B	B	B	B	B	"	A	B	H	L	L	L	H	"	"			
			73	A	A	B	B	B	B	B	"	A	B	H	L	L	H	H	"	"			
			74	A	B 6/	B	B	B	B	B	"	A	B	H	L	L	H	H	"	"			
			75	A	A	B	B	B	B	B	"	A	B	H	L	H	L	H	"	"			
			76	A	B	B	B	B	B	A	"	A	B	H	L	H	L	H	"	"			
			77	A	A	B	B	B	B	A	"	A	B	H	L	H	L	H	"	"			
			78	A	B	B	B	B	B	A	"	A	B	H	L	H	L	H	"	"			
			79	A	A	B	B	B	B	B	"	A	B	H	L	H	H	H	"	"			
			80	A	B	B	B	B	B	B	"	A	B	H	L	H	H	H	"	"			
			81	A	A	B	B	B	B	B	"	A	B	H	H	L	L	H	"	"			
			82	A	B	B	B	B	B	B	"	A	B	H	H	L	L	H	"	"			
			83	A	A	B	B	B	B	B	"	A	B	H	H	L	L	H	"	"			
			84	A	B	B	B	B	B	B	"	A	B	H	H	L	H	H	"	"			
			85	A	A	B	B	B	B	B	"	A	B	H	H	L	H	L	"	"			
			86	A	B	B	B	B	B	B	"	A	B	H	H	H	L	L	"	"			
			87	A	A	B	B	B	B	B	"	A	B	H	H	H	H	L	"	"			
			88	A	B	B	B	B	B	B	"	A	B	H	H	H	H	L	"	"			
			89	A	B	B	B	B	B	B	"	A	A	H	H	H	H	H	"	"			
			90	A	A	B	B	B	B	B	"	A	A	H	H	H	H	H	"	"			
			91	A	B	B	B	B	B	B	"	A	B	H	H	H	H	L	"	"			
			92	A	A	B	B	B	B	B	"	A	B	L	L	L	L	H	"	"			
			93	B	B	A	A	A	A	B	"	B	B	L	L	L	L	L	"	"			
			94	B	A	A	A	A	A	A	"	B	B	H	H	H	H	H	"	"			
			95	B	B	A	A	A	A	A	"	A	B	H	H	H	H	L	"	"			
			96	B	A	A	A	A	A	A	"	A	B	H	H	H	H	L	"	"			
			97	B	B	A	A	A	A	A	"	A	B	H	H	H	L	L	"	"			
			98	B	A	A	A	A	A	A	"	A	B	H	H	L	L	L	"	"			
			99	B	B	A 6/	A	A	A	A	"	A	B	H	H	L	H	L	"	"			
			100	B	A	A	A	A	A	A	"	A	B	H	H	L	L	L	"	"			
			101	B	B	A	A	A	A	A	"	A	B	H	H	L	L	L	"	"			
			102	B	A	A	A	A	A	A	"	A	B	H	H	L	L	H	"	"			
			103	B	B	A	A	A	A	A	"	A	B	H	H	L	H	H	"	"			
			104	B	A	A	A	A	A	A	"	A	B	H	H	L	H	H	"	"			
			105	B	B	A	A	A	A	B	"	A	B	H	H	H	L	L	"	"			
			106	B	A	A	A	A	A	B	"	A	B	H	H	H	L	L	"	"			
			107	B	B	A	A	A	A	B	"	A	B	H	H	H	L	L	"	"			
			108	B	A	A	A	A	A	B	"	A	B	H	L	L	H	L	"	"			

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				U/D	CLK	A	B	C	D	$\overline{ENP}$	GND	$\overline{LOAD}$	$\overline{ENT}$	QD	QC	QB	QA	$\overline{RCO}$	V_{CC}				
7 T_C = +25°C	Truth table test 5/	3014 Fig. 3	109	B 6/	B	A 6/	A	A	A	B	GND	A	B	H	L	L	H	H	5.0 V	8/			
			110	B	A	A	A	A	A	B	"	A	B	H	L	L	L	H	"	"			
			111	B	B	A	A	A	A	B	"	A	B	H	L	L	L	H	"	"			
			112	B	A	A	A	A	A	B	"	A	B	L	H	H	H	H	"	"			
			113	B	B	A	A	A	A	B	"	A	B	L	H	H	H	H	"	"			
			114	B	A	A	A	A	A	B	"	A	B	L	H	H	L	H	"	"			
			115	B	B	A	A	A	A	B	"	A	B	L	H	H	L	H	"	"			
			116	B	A	A	A	A	A	B	"	A	B	L	H	L	H	H	"	"			
			117	B	B	A	A	A	A	B	"	A	B	L	H	L	H	H	"	"			
			118	B	A	A	A	A	A	B	"	A	B	L	H	L	L	H	"	"			
			119	B	B	A	A	A	A	B	"	A	B	L	H	L	L	H	"	"			
			120	B	A	A	A	A	A	B	"	A	B	L	L	H	H	H	"	"			
			121	B	B	A	A	A	A	B	"	A	B	L	L	H	H	H	"	"			
			122	B	A	A	A	A	A	B	"	A	B	L	L	H	L	H	"	"			
			123	B	B	A	A	A	A	B	"	A	B	L	L	H	L	H	"	"			
			124	B	A	A	A	A	A	B	"	A	B	L	L	L	H	H	"	"			
			125	B	B	A	A	A	A	B	"	A	B	L	L	L	H	H	"	"			
			126	B	A	A	A	A	A	B	"	A	B	L	L	L	L	H	"	"			
			127	B	B	A	A	A	A	B	"	A	A	L	L	L	L	H	"	"			
			128	B	A	A	A	A	A	B	"	A	A	L	L	L	L	H	"	"			
			129	B	B	A	A	A	A	B	"	A	B	L	L	L	L	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 8/	3003 Fig. 8	130	4.5 V	IN					GND	GND	4.5 V	GND				OUT		5.0 V	QA	30		MHz
			131	GND	"					"	"	4.5 V	"				"		"	QA	30		MHz
	t_{PLH1}		132	4.5 V	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	GND	"	OUT	OUT	OUT	"		"	CLK to QA	5	16	ns
			133	"	"	"				"	"	"	"						"	CLK to QB	"	"	"
			134	"	"	"				"	"	"	"						"	CLK to QC	"	"	"
			135	"	"	"				"	"	"	"						"	CLK to QD	"	"	"
	t_{PHL1}		136	"	"	GND	GND	GND	GND	"	"	"	"	OUT	OUT	OUT	OUT		"	CLK to QA	"	"	"
			137	"	"	"				"	"	"	"						"	CLK to QB	"	"	"
			138	"	"	"				"	"	"	"						"	CLK to QC	"	"	"
			139	"	"	"				"	"	"	"						"	CLK to QD	"	"	"
	t_{PLH2}		140	"	"	GND	GND	GND	GND	"	"	"	"					OUT	"	CLK to $\overline{RCO}$	6	20	"
	t_{PHL2}		141	"	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"					"	"	CLK to $\overline{RCO}$	6	20	"
	t_{PLH8}		142	"	2/	"	"	"	"	"	"	"	IN					"	"	$\overline{ENT}$ to $\overline{RCO}$	5	16	"
	t_{PHL8}		143	"	"	"	"	"	"	"	"	"	IN					"	"	$\overline{ENT}$ to $\overline{RCO}$	3	13	"
	t_{PLH11}		144	IN	"	GND	GND	GND	GND	"	"	"						"	"	U/$\overline{D}$ to $\overline{RCO}$	5	23	"
	t_{PHL11}		145	IN	"	GND	GND	GND	GND	"	"	"						"	"	U/$\overline{D}$ to $\overline{RCO}$	5	19	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 U/D	2 3 CLK	3 4 A	4 5 B	5 7 C	6 8 D	7 9 ENP	8 10 GND	9 12 LOAD	10 13 ENT	11 14 QD	12 15 QC	13 17 QB	14 18 QA	15 19 RCO	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = +125°C	f_{MAX} 8/	3003 Fig. 6																			25		MHz
	t_{PLH1}																				5	19	ns
	t_{PHL1}																				5	20	"
	t_{PLH2}		Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																		10	32	"
	t_{PHL2}																				6	25	"
	t_{PLH8}																				5	19	"
	t_{PHL8}																				3	19	"
	t_{PLH11}																				5	25	"
	t_{PHL11}																				5	25	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Case 2 pins not referenced are N/C.

2/ Apply one clock pulse prior to test as follows: ___|‾|___ 2.5 V min/5.5 V max, 0 V.

3/ I_{IL} limits shall be as follows:

Test	Min/max limits in µA for circuit A	B	C
I_{IL}	0/-200		

4/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.

5/ Only a summary of attributes data is required.

6/ A = 3.0 V minimum; B = 0.0 V or GND.

7/ H >1.5 V; L <1.5 V.

8/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 $\overline{ALOAD}$	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 $\overline{ACLR}$	9 $\overline{SCLR}$	10 GND	11 $\overline{SLOAD}$	12 ENT	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 CCO	19 RCO	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	1/	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	0.8 V	2.0 V					0.8 V		-400 mA	4.5 V	RCO	2.5		V
			2	2.0 V	1/	"	2.0 V	2.0 V	2.0 V	2.0 V	"	"	"	"	2.0 V					"	-400 mA		"	CCO	2.5		"
	V_{OH2}		3	0.8 V	"	"				0.8 V	"	"	"	"	0.8 V				-1 mA				"	QA	2.4		"
			4	"	"		2.0 V			"	"	"	"	"	"				"				"	QB	"		"
			5	"	"			2.0 V		"	"	"	"	"	"	-1 mA	-1 mA	-1 mA	"				"	QC	"		"
			6	"	"				2.0 V	"	"	"	"	"	"				"				"	QD	"		"
	V_{OL1}	3007	7	2.0 V	"	0.8 V	0.8 V	0.8 V	0.8 V	"	"	"	"	"	"					"		4 mA	"	RCO		0.4	"
			8	2.0 V	"	"	"	"	"	"	"	"	"	"	"					"	4 mA		"	CCO		"	"
	V_{OL2}		9	0.8 V	"	"				"	"	"	"	"	"					"			"	QA		"	"
			10	"	"		0.8 V			"	"	"	"	"	"				12 mA	"			"	QB		"	"
			11	"	"			0.8 V		"	"	"	"	"	"			12 mA		"			"	QC		"	"
			12	"	"				0.8 V	"	"	"	"	"	"	12 mA	12 mA			"			"	QD		"	"
	V_{IC}		13	-18 mA																				$\overline{ALOAD}$		-1.5	V
			14		-18 mA																			CLK		"	"
			15			-18 mA																		A		"	"
			16				-18 mA																	B		"	"
			17					-18 mA																C		"	"
			18						-18 mA															D		"	"
			19							-18 mA														ENP		"	"
			20								-18 mA													$\overline{ACLR}$		"	"
			21									-18 mA												$\overline{SCLR}$		"	"
			22											-18 mA										$\overline{SLOAD}$		"	"
			23												-18 mA									ENT		"	"
			24																	-18 mA				$\overline{G}$		"	"
	I_{IL}	3009	25	0.4 V																			5.5 V	$\overline{ALOAD}$	2/	2/	µA
			26		0.4 V																		"	CLK	"	"	"
			27			0.4 V																		A	"	"	"
			28				0.4 V																	B	"	"	"
			29					0.4 V																C	"	"	"
			30						0.4 V															D	"	"	"
			31							0.4 V														ENP	"	"	"
			32								0.4 V													$\overline{ACLR}$	"	"	"
			33									0.4 V												$\overline{SCLR}$	"	"	"
			34											0.4 V										$\overline{SLOAD}$	"	"	"
			35												0.4 V									ENT	"	"	"
			36																	0.4 V				$\overline{G}$	"	"	"
	I_{IH1}	3010	37							2.7 V														ENP		40	µA
			38												2.7 V									ENT		40	"
			39	2.7 V																				$\overline{ALOAD}$		20	"
			40		2.7 V																			CLK		"	"
			41			2.7 V																		A		"	"
			42				2.7 V																	B		"	"
			43					2.7 V																C		"	"
			44						2.7 V															D		"	"
			45								2.7 V													$\overline{ACLR}$		"	"
			46									2.7 V												$\overline{SCLR}$		"	"
			47											2.7 V										$\overline{SLOAD}$		"	"
			48																	2.7 V				$\overline{G}$		"	"

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{ALOAD}$	CLK	A	B	C	D	ENP	$\overline{ACLR}$	$\overline{SCLR}$	GND	$\overline{SLOAD}$	ENT	QD	QC	QB	QA	$\overline{G}$	CCO	RCO	V_{CC}				
1 T_C = +25°C	I_{IH2}	3010	49							7.0 V			GND										5.5 V	ENP		200	µA
			50										"		7.0 V								"	ENT		200	"
			51	7.0 V									"										"	$\overline{ALOAD}$		100	"
			52		7.0 V								"										"	CLK		"	"
			53			7.0 V							"										"	A		"	"
			54				7.0 V						"										"	B		"	"
			55					7.0 V					"										"	C		"	"
			56						7.0 V				"										"	D		"	"
			57								7.0 V		"										"	$\overline{ACLR}$		"	"
			58									7.0 V	"										"	$\overline{SCLR}$		"	"
			59										"	7.0 V									"	$\overline{SLOAD}$		"	"
			60										"							7.0 V			"	$\overline{G}$		"	"
	I_O 3/	3011	61	5.5 V	1/	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	GND	5.5 V					GND		2.25 V		RCO	-15	-70	mA
			62	"	"	"	5.5 V	5.5 V	5.5 V	"	"	"	"	"	"					"	2.25 V		"	CCO	-15	-70	"
			63	"	"					"	"	"	"	"	"				2.25 V	"			"	QA	-30	-112	"
			64	"	"		5.5 V			"	"	"	"	"	"			2.25 V		"			"	QB	"	"	"
			65	"	"			5.5 V		"	"	"	"	"	"		2.25 V			"			"	QC	"	"	"
			66	"	"				5.5 V	"	"	"	"	"	"	2.25 V				"			"	QD	"	"	"
	I_{OZL}		67	2.0 V	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	0.8 V	2.0 V					2.0 V				QA		-20	µA
			68	"	"	"	"	"	"	"	"	"	"	"	"			0.4 V		"				QB		"	"
			69	"	"	"	"	"	"	"	"	"	"	"	"		0.4 V		0.4 V	"				QC		"	"
			70	"	"	"	"	"	"	"	"	"	"	"	"	0.4 V				"				QD		"	"
	I_{OZH}		71	"	"	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"	"	"	"	0.8 V				2.7 V	"				QA		20	"
			72	"	"	"	"	"	"	"	"	"	"	"	"			2.7 V		"			"	QB		"	"
			73	"	"	"	"	"	"	"	"	"	"	"	"		2.7 V			"			"	QC		"	"
			74	"	"	"	"	"	"	"	"	"	"	"	"	2.7 V				"			"	QD		"	"
	I_{CCH}	3005	75	4.5 V	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	GND	4.5 V					GND			"	V_{CC}		27	mA
	I_{CCL}		76	GND	"	GND	GND	GND	GND	GND	GND	GND	"	"	GND					GND			"	"		33	"
	I_{CCZ}		77	GND	"	GND	GND	GND	GND	GND	4.5 V	4.5 V	"	"	GND					4.5 V			"	"		36	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 T_C = +25°C	Truth table test 4/	3014 Fig. 3	78	A 5/	B 5/	A	B	B	A	A	B	A	GND	A	A	L	L	L	L	B	L	L	4.5 V	6/			
			79	B	B	A	B	B	A	A	A	A	"	A	A	H	L	H	L	B	L	L	"	"			
			80	A	A	A	B	B	A	A	A	A	"	A	A	H	L	H	H	B	L	L	"	"			
			81	A	B	A	B	B	A	A	A	A	"	A	A	H	L	H	L	B	L	L	"	"			
			82	A	A	A	B	B	A	A	A	A	"	A	A	H	L	H	H	B	L	L	"	"			
			83	A	B	A	B	B	A	A	A	A	"	A	A	H	L	H	H	B	L	L	"	"			
			84	A	A	A	B	B	A	A	A	A	"	A	A	H	H	L	L	B	L	L	"	"			
			85	A	B	A	B	B	A	A	A	A	"	A	A	H	H	L	L	B	L	L	"	"			
			86	A	A	A	B	B	A	A	A	A	"	A	A	H	H	L	H	B	L	L	"	"			
			87	A	B	A	B	B	A	A	A	A	"	A	A	H	H	L	H	B	L	L	"	"			
			88	A	A	A	B	B	A	A	A	A	"	A	A	H	H	H	L	B	L	L	"	"			
			89	A	B	A	B	B	A	A	A	A	"	A	A	H	H	H	L	B	L	L	"	"			
			90	A	A	A	B	B	A	A	A	A	"	A	A	H	H	H	H	B	L	L	"	"			
			91	A	B	A	B	B	A	A	A	A	"	A	B	H	H	H	H	B	H	H	"	"			
			92	A	A	A	B	B	A	A	A	A	"	A	B	H	H	H	H	B	L	L	"	"			
			93	A	B	A	B	B	A	A	A	A	"	A	B	H	H	H	H	B	L	L	"	"			
			94	A	B	A	B	B	A	A	A	A	"	A	A	H	H	H	H	B	H	H	"	"			
			95	A	A	A	B	B	A	A	A	A	"	A	A	L	L	L	L	B	L	L	"	"			
			96	A	B	A	B	B	A	A	A	A	"	A	A	L	L	L	L	B	L	L	"	"			
			97	A	A	A	B	B	A	A	A	A	"	A	A	L	L	L	H	B	L	L	"	"			

See footnotes at end of devic type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 ALOAD	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 ACLR	9 SCLR	10 GND	11 SLOAD	12 ENT	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 CCO	19 RCO	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = +25°C	Truth table test 4/	3014 Fig. 3	98	A 5/	B 5/	A	B	B	A	A	A	A	GND	A	A	L	L	L	H	B	L	L	4.5 V	6/			
			99	A	A	A	B	B	A	A	A	A	"	A	A	L	L	H	L	B	L	L	"	"			
			100	A	B	A	B	B	A	A	A	A	"	A	A	L	L	H	L	B	L	L	"	"			
			101	A	A	A	B	B	A	A	A	A	"	A	A	L	L	H	H	B	L	L	"	"			
			102	A	B	A	B	B	A	B	A	A	"	A	A	L	L	H	H	B	L	L	"	"			
			103	A	A	A	B	B	A	B	A	A	"	A	A	L	L	H	H	B	L	L	"	"			
			104	A	B	A	B	B	A	A	A	A	"	A	A	L	L	H	H	B	L	L	"	"			
			105	A	A	A	B	B	A	A	A	A	"	A	A	L	H	L	L	B	L	L	"	"			
			106	A	B	A	B	B	A	A	A	A	"	A	A	L	H	L	L	B	L	L	"	"			
			107	A	A	A	B	B	A	A	A	A	"	A	A	L	H	L	H	B	L	L	"	"			
			108	A	B	A	B	B	A	A	A	A	"	A	A	L	H	L	H	B	L	L	"	"			
			109	A	A	A	B	B	A	A	A	A	"	A	A	L	H	H	L	B	L	L	"	"			
			110	A	B	A	B	B	A	A	A	B	"	A	A	L	H	H	L	B	L	L	"	"			
			111	A	A	A	B	B	A	A	A	B	"	A	A	L	L	L	L	B	L	L	"	"			
			112	A	B	A	B	B	A	A	A	A	"	B	A	L	L	L	L	B	L	L	"	"			
			113	A	A	A	B	B	A	A	A	A	"	B	A	H	L	H	L	B	L	L	"	"			
			114	A	B	A	B	B	A	A	A	A	"	A	A	H	L	H	L	B	L	L	"	"			
			115	A	A	A	B	B	A	A	A	A	"	A	A	H	L	H	H	B	L	L	"	"			
			116	A	B	A	B	B	A	A	A	A	"	A	A	H	L	H	H	B	L	L	"	"			
			117	A	B	A	B	B	A	A	B	A	"	A	A	L	L	L	L	B	L	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																										
9 T_C = +25°C	f_{MAX} 8/	3003 Fig. 9	118	4.5 V	IN					4.5 V	4.5 V	4.5 V	GND	4.5 V	4.5 V				OUT	GND			5.0 V	QA	25		MHz
	t_{PLH1}		119	7/	"	GND	GND	GND	GND	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	CLK to QA	4	12	ns
			120	"	"					"	"	"	"	"	"					"			"	CLK to QB	"	"	"
			121	"	"					"	"	"	"	"	"					"			"	CLK to QC	"	"	"
			122	"	"					"	"	"	"	"	"					"			"	CLK to QD	"	"	"
	t_{PHL1}		123	"	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	CLK to QA	5	18	"
			124	"	"					"	"	"	"	"	"					"			"	CLK to QB	"	"	"
			125	"	"					"	"	"	"	"	"					"			"	CLK to QC	"	"	"
			126	"	"					"	"	"	"	"	"					"			"	CLK to QD	"	"	"
	t_{PLH2}		127	"	"	GND	4.5 V	4.5 V	"	"	"	"	"	"	"					"		OUT	"	CLK to RCO	9	29	"
	t_{PHL2}		128	"	"	4.5 V	"	"	"	"	"	"	"	"	"					"		OUT	"	CLK to RCO	8	24	"
	t_{PLH3}		129	"	"	"	"	"	"	"	"	"	"	"	"					"	OUT		"	CLK to CCO	8	26	"
	t_{PHL3}		130	"	"	"	"	"	"	"	"	"	"	"	"					"	OUT		"	CLK to CCO	5	16	"
	t_{PLH4}		131	IN		"	4.5 V	4.5 V	4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{ALOAD}$ to QA	10	35	"
			132	"						"	"	"	"	"	"					"			"	$\overline{ALOAD}$ to QB	"	"	"
			133	"						"	"	"	"	"	"					"			"	$\overline{ALOAD}$ to QC	"	"	"
			134	"						"	"	"	"	"	"					"			"	$\overline{ALOAD}$ to QD	"	"	"
	t_{PHL4}		135	"		GND	GND	GND	GND	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{ALOAD}$ to QA	7	23	"
			136	"						"	"	"	"	"	"					"			"	$\overline{ALOAD}$ to QB	"	"	"
			137	"						"	"	"	"	"	"					"			"	$\overline{ALOAD}$ to QC	"	"	"
			138	"						"	"	"	"	"	"					"			"	$\overline{ALOAD}$ to QD	"	"	"
	t_{PLH5}		139	"		4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	"	"					"		OUT	"	$\overline{ALOAD}$ to RCO	15	42	"
	t_{PHL5}		140	"		GND	GND	GND	GND	"	"	"	"	"	"					"		OUT	"	$\overline{ALOAD}$ to RCO	12	30	"

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 $\overline{ALOAD}$	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 $\overline{ACLR}$	9 $\overline{SCLR}$	10 GND	11 $\overline{SLOAD}$	12 ENT	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 CCO	19 RCO	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PLH6}	3003 Fig. 9	141	IN	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	GND	4.5 V	4.5 V					GND	OUT		5.0 V	$\overline{ALOAD}$ to CCO	25	55	ns
	t_{PHL6}		142	IN	GND	GND	GND	GND	GND	"	"	"	"	"	"					"	OUT		"	$\overline{ALOAD}$ to CCO	12	33	"
	t_{PLH7}		143	GND		IN				"	"	"	"	"	"					"			"	A to QA	8	30	"
			144	"			IN			"	"	"	"	"	"					"			"	B to QB	"	"	"
			145	"				IN		"	"	"	"	"	"					"			"	C to QC	"	"	"
			146	"					IN	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	D to QD	"	"	"
	t_{PHL7}		147	"		IN				"	"	"	"	"	"					"			"	A to QA	7	22	"
			148	"			IN			"	"	"	"	"	"					"			"	B to QB	"	"	"
			149	"				IN		"	"	"	"	"	"					"			"	C to QC	"	"	"
			150	"					IN	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	D to QD	"	"	"
	t_{PLH8}		151	7/		4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	"	IN					"		OUT	"	ENT to RCO	5	16	"
	t_{PHL8}		152	"		"	"	"	"	"	"	"	"	"	"					"		OUT	"	ENT to RCO	4	14	"
	t_{PLH9}		153	"		"	"	"	"	"	"	"	"	"	"					"	OUT		"	ENT to CCO	12	32	"
	t_{PHL9}		154	"		"	"	"	"	"	"	"	"	"	"					"	"		"	ENT to CCO	4	12	"
	t_{PLH10}		155	"		"	"	"	"	IN	"	"	"	"	4.5 V					"	"		"	ENP to CCO	5	18	"
	t_{PHL10}		156	"		"	"	"	"	IN	"	"	"	"	"					"	"		"	ENP to CCO	4	12	"
	t_{PHL12}		157	"		"				4.5 V	IN	"	"	"	"					"			"	$\overline{ACLR}$ to QA	7	22	"
			158	"			4.5 V			"	"	"	"	"	"					"			"	$\overline{ACLR}$ to QB	"	"	"
			159	"				4.5 V		"	"	"	"	"	"					"			"	$\overline{ACLR}$ to QC	"	"	"
			160	"					4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{ACLR}$ to QD	"	"	"
	t_{PZH}		161	"		4.5 V				"	4.5 V	"	"	"	"					IN			"	$\overline{G}$ to QA	5	19	"
			162	"			4.5 V			"	"	"	"	"	"					"			"	$\overline{G}$ to QB	"	"	"
			163	"				4.5 V		"	"	"	"	"	"					"			"	$\overline{G}$ to QC	"	"	"
			164	"					4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{G}$ to QD	"	"	"
	t_{PZL}		165	"		GND				"	"	"	"	"	"					"			"	$\overline{G}$ to QA	8	23	"
			166	"			GND			"	"	"	"	"	"					"			"	$\overline{G}$ to QB	"	"	"
			167	"				GND		"	"	"	"	"	"					"			"	$\overline{G}$ to QC	"	"	"
			168	"					GND	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{G}$ to QD	"	"	"
	t_{PHZ}		169	"		4.5 V				"	"	"	"	"	"					"			"	$\overline{G}$ to QA	2	10	"
			170	"			4.5 V			"	"	"	"	"	"					"			"	$\overline{G}$ to QB	"	"	"
			171	"				4.5 V		"	"	"	"	"	"					"			"	$\overline{G}$ to QC	"	"	"
			172	"					4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{G}$ to QD	"	"	"
	t_{PLZ}		173	"		GND				"	"	"	"	"	"					"			"	$\overline{G}$ to QA	4	15	"
			174	"			GND			"	"	"	"	"	"					"			"	$\overline{G}$ to QB	"	"	"
			175	"				GND		"	"	"	"	"	"					"			"	$\overline{G}$ to QC	"	"	"
			176	"					GND	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{G}$ to QD	"	"	"

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 $\overline{ALOAD}$	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 $\overline{ACLR}$	9 $\overline{SCLR}$	10 GND	11 $\overline{SLOAD}$	12 ENT	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 CCO	19 RCO	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10 T_C = +125°C	f_{MAX}	3003 Fig. 9																							25		MHz
	t_{PLH1}																								4	15	ns
	t_{PHL1}																								5	21	"
	t_{PLH2}																								9	35	"
	t_{PHL2}																								8	29	"
	t_{PLH3}																								8	31	"
	t_{PHL3}																								5	20	"
	t_{PLH4}																								10	38	"
	t_{PHL4}																								7	27	"
	t_{PLH5}																								15	55	"
	t_{PHL5}		Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																						12	35	"
	t_{PLH6}																								25	65	"
	t_{PHL6}																								12	42	"
	t_{PLH7}																								8	35	"
	t_{PHL7}																								7	27	"
	t_{PLH8}																								5	20	"
	t_{PHL8}																								4	18	"
	t_{PLH9}																								12	35	"
	t_{PHL9}																								4	15	"
	t_{PLH10}																								5	22	"

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 ALOAD	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 $\overline{ACLR}$	9 $\overline{SCLR}$	10 GND	11 $\overline{SLOAD}$	12 ENT	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 CCO	19 RCO	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10 T_C = +125°C	t_{PHL10}	3003 Fig. 9	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																						4	14	ns
	t_{PHL12}																								7	28	"
	t_{PZH}																								5	24	"
	t_{PZL}																								8	28	"
	t_{PHZ}																								2	15	"
	t_{PLZ}																								4	20	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

1/ Apply one clock pulse prior to test as follows: ___|‾|___ 2.5 V min/5.5 V max, 0 V.

2/ I_{IL} limits shall be as follows:

Test	Device type	Min/max limits in µA for circuit A	B	C
I_{IL}	01	0/-200		

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.

4/ Only a summary of attributes data is required.

5/ A = 3.0 V minimum; B = 0.0 V or GND.

6/ H >1.5 V; L <1.5 V.

7/ Apply one clock pulse prior to test as follows: ___|‾|___ 2.5 V min/5.5 V max, 0 V.

8/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	U/D	CLK	A	B	C	D	ENP	ACLR	SCLR	GND	SLOAD	ENT	QD	QC	QB	QA	G	CCO	RCO	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	1/	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.0 V	2.0 V	GND	0.8 V	0.8 V					0.8 V	-400 µA	-400 µA	4.5 V	RCO	2.5		V
			2	"		0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.0 V	2.0 V	"	"	"					"			"	CCO	2.5		"
	V_{OH2}		3	"	"	2.0 V				"	"	"	"	"	"					"			"	QA	2.4		"
			4	"	"		2.0 V			"	"	"	"	"	"				-1 mA	"			"	QB	"		"
			5	"	"			2.0 V		"	"	"	"	"	"		-1 mA	-1 mA		"			"	QC	"		"
			6	"	"				2.0 V	"	"	"	"	"	"	-1 mA				"			"	QD	"		"
	V_{OL1}	3007	7	"	"	2.0 V	2.0 V	2.0 V	"	"	"	"	"	"	"					"		4 mA	"	RCO		0.4	"
			8	"	"	2.0 V	2.0 V	2.0 V	"	"	"	"	"	"	"					"	4 mA		"	CCO		"	"
	V_{OL2}		9	"	"	0.8 V				"	"	"	"	"	"					"			"	QA		"	"
			10	"	"		0.8 V			"	"	"	"	"	"		12 mA	12 mA	12 mA	"			"	QB		"	"
			11	"	"			0.8 V		"	"	"	"	"	"					"			"	QC		"	"
			12	"	"				0.8 V	"	"	"	"	"	"	12 mA				"			"	QD		"	"
	V_{IC}		13	-18 mA									"											U/D		-1.5	V
			14		-18 mA								"											CLK		"	"
			15			-18 mA							"											A		"	"
			16				-18 mA						"											B		"	"
			17					-18 mA					"											C		"	"
			18						-18 mA				"											D		"	"
			19							-18 mA			"											ENP		"	"
			20								-18 mA		"											ACLR		"	"
			21									-18 mA	"											SCLR		"	"
			22										"	-18 mA										SLOAD		"	"
			23										"		-18 mA									ENT		"	"
			24										"							-18 mA				G		"	"
	I_{IL}	3009	25	0.4 V									"										5.5 V	U/D	2/	2/	µA
			26		0.4 V								"										"	CLK	"	"	"
			27			0.4 V							"										"	A	"	"	"
			28				0.4 V						"										"	B	"	"	"
			29					0.4 V					"										"	C	"	"	"
			30						0.4 V				"										"	D	"	"	"
			31							0.4 V			"										"	ENP	"	"	"
			32								0.4 V		"										"	ACLR	"	"	"
			33									0.4 V	"										"	SCLR	"	"	"
			34										"	0.4 V									"	SLOAD	"	"	"
			35										"		0.4 V								"	ENT	"	"	"
			36										"							0.4 V				G	"	"	"
	I_{IH1}	3010	37	2.7 V									"										"	U/D		20	"
			38		2.7 V								"										"	CLK		"	"
			39			2.7 V							"										"	A		"	"
			40				2.7 V						"										"	B		"	"
			41					2.7 V					"										"	C		"	"
			42						2.7 V				"										"	D		"	"
			43							2.7 V			"										"	ENP		"	"
			44								2.7 V		"										"	ACLR		"	"
			45									2.7 V	"										"	SCLR		"	"
			46										"	2.7 V									"	SLOAD		"	"
			47										"		2.7 V								"	ENT		"	"
			48										"							2.7 V				G		"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 U/D	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 ACLR	9 SCLR	10 GND	11 SLOAD	12 ENT	13 QD	14 QC	15 QB	16 QA	17 G	18 CCO	19 RCO	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH2}	3010	49	7.0 V									GND									5.5 V	U/D		100	µA
			50		7.0 V								"									"	CLK		"	"
			51			7.0 V							"									"	A		"	"
			52				7.0 V						"									"	B		"	"
			53					7.0 V					"									"	C		"	"
			54						7.0 V				"									"	D		"	"
			55							7.0 V			"									"	ENP		"	"
			56								7.0 V		"									"	ACLR		"	"
			57									7.0 V	"									"	SCLR		"	"
			58										"	7.0 V								"	SLOAD		"	"
			59										"		7.0 V							"	ENT		"	"
			60										"						7.0 V			"	G		"	"
	I_O 3/	3011	61	5.5 V	1/	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	"	GND	5.5 V					GND	2.25 V	2.25 V	"	RCO	-15	-70	mA
			62	"	"	GND	GND	GND	GND	"	"	"	"	"	"					"	"	"	"	CCO	-15	-70	"
			63	"	"	5.5 V				"	"	"	"	"	"				2.25 V	"			"	QA	-30	-112	"
			64	"	"		5.5 V			"	"	"	"	"	"			2.25 V		"			"	QB	"	"	"
			65	"	"			5.5 V		"	"	"	"	"	"		2.25 V			"			"	QC	"	"	"
			66	"	"				5.5 V	"	"	"	"	"	"	2.25 V				"			"	QD	"	"	"
	I_{OZL}		67	2.0 V	"	2.0 V				2.0 V	2.0 V	2.0 V	"	0.8 V	2.0 V				0.4 V	2.0 V			"	QA		-20	µA
			68	"	"		2.0 V			"	"	"	"	"	"			0.4 V		"			"	QB		"	"
			69	"	"			2.0 V		"	"	"	"	"	"		0.4 V			"			"	QC		"	"
			70	"	"				2.0 V	"	"	"	"	"	"	0.4 V				"			"	QD		"	"
	I_{OZH}		71	"	"	0.8 V				"	"	"	"	"	"				2.7 V	"			"	QA		20	"
			72	"	"		0.8 V			"	"	"	"	"	"			2.7 V		"			"	QB		"	"
			73	"	"			0.8 V		"	"	"	"	"	"		2.7 V			"			"	QC		"	"
			74	"	"				0.8 V	"	"	"	"	"	"	2.7 V				"			"	QD		"	"
	I_{CCH}	3005	75	4.5 V	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	GND	4.5 V								"	V_{CC}		26	mA
	I_{CCL}		76	GND	"	GND	GND	GND	GND	GND	GND	GND	"	"	GND								"	V_{CC}		32	"
	I_{CCZ}		77	GND	"	GND	GND	GND	GND	GND	4.5 V	4.5 V	"	"	GND								"	V_{CC}		32	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																									
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																									
7 T_C = +25°C	Truth table test 4/	3014 Fig. 3	78	A 5/	B 5/	B	B	B	B	B	B	B	GND	A	B	L	L	L	L	B	H	H	4.5 V	6/			
			79	A	B	B	B	B	B	B	B	B	"	B	B	L	L	L	L	B	H	H	"				
			80	A	B	B	B	B	B	B	B	B	"	A	B	L	L	L	L	B	H	H	"				
			81	A	B	B	B	B	B	B	B	B	"	A	B	L	L	L	H	B	H	H	"				
			82	A	B	B	B	B	B	B	B	B	"	A	B	L	L	L	H	B	H	H	"				
			83	A	B	B	B	B	B	B	B	B	"	A	B	L	L	H	L	B	H	H	"				
			84	A	B	B	B	B	B	B	B	B	"	A	B	L	L	H	L	B	H	H	"				
			85	A	B	B	B	B	B	B	B	B	"	A	B	L	L	H	H	B	H	H	"				
			86	A	B	B	B	B	B	B	B	B	"	A	B	L	L	H	H	B	H	H	"				
			87	A	B	B	B	B	B	B	B	B	"	A	B	L	H	L	L	B	H	H	"				
			88	A	B	B	B	B	B	B	B	B	"	A	B	L	H	L	L	B	H	H	"				
			89	A	B	B	B	B	B	B	B	B	"	A	B	L	H	L	H	B	H	H	"				
			90	A	B	B	B	B	B	B	B	B	"	A	B	L	H	L	H	B	H	H	"				
			91	A	B	B	B	B	B	B	B	B	"	A	B	L	H	H	L	B	H	H	"				
			92	A	B	B	B	B	B	B	B	B	"	A	B	L	H	H	L	B	H	H	"				
			93	A	B	B	B	B	B	B	B	B	"	A	B	L	H	H	H	B	H	H	"				
			94	A	B	B	B	B	B	B	B	B	"	A	B	L	H	H	H	B	H	H	"				
			95	A	B	B	B	B	B	B	B	B	"	A	B	L	H	H	H	B	H	H	"				
			96	A	B	B	B	B	B	B	B	B	"	A	B	L	H	H	H	B	H	H	"				
			97	A	B	B	B	B	B	B	B	B	"	A	B	L	H	H	H	B	H	H	"				

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 U/D	2 CLK	3 A	4 B	5 C	6 D	7 $\overline{ENP}$	8 $\overline{ACLR}$	9 $\overline{SCLR}$	10 GND	11 $\overline{SLOAD}$	12 $\overline{ENT}$	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 $\overline{CCO}$	19 $\overline{RCO}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = +25°C	Truth table test 4/	3015 Fig. 3	98	A 5/	A 5/	B	B	B	B	B	A	A	GND	A	B	H	L	L	H	B	H	H	4.5 V	6/			
			99	A	B	B	B	B	B	B	A	A	"	A	B	H	L	L	H	B	H	H	"	"			
			100	A	A	B	B	B	B	B	A	A	"	A	B	H	L	H	L	B	H	H	"	"			
			101	A	B	B	B	B	B	B	A	A	"	A	B	H	L	H	L	B	H	H	"	"			
			102	A	A	B	B	B	B	A	A	A	"	A	B	H	L	H	L	B	H	H	"	"			
			103	A	B	B	B	B	B	B	A	A	"	A	B	H	L	H	L	B	H	H	"	"			
			104	A	A	B	B	B	B	B	A	A	"	A	B	H	L	H	H	B	H	H	"	"			
			105	A	B	B	B	B	B	B	A	A	"	A	B	H	L	H	H	B	H	H	"	"			
			106	A	A	B	B	B	B	B	A	A	"	A	B	H	H	L	L	B	H	H	"	"			
			107	A	B	B	B	B	B	B	A	A	"	A	B	H	H	L	L	B	H	H	"	"			
			108	A	A	B	B	B	B	B	A	A	"	A	B	H	H	L	H	B	H	H	"	"			
			109	A	B	B	B	B	B	B	A	A	"	A	B	H	H	L	H	B	H	H	"	"			
			110	A	A	B	B	B	B	B	A	A	"	A	B	H	H	H	L	B	H	H	"	"			
			111	A	B	B	B	B	B	B	A	A	"	A	B	H	H	H	L	B	H	H	"	"			
			112	A	A	B	B	B	B	B	A	A	"	A	B	H	H	H	H	B	H	L	"	"			
			113	A	B	B	B	B	B	B	A	A	"	A	B	H	H	H	H	B	L	L	"	"			
			114	A	B	B	B	B	B	B	A	A	"	A	A	H	H	H	H	B	H	H	"	"			
			115	A	A	B	B	B	B	B	A	A	"	A	A	H	H	H	H	B	H	H	"	"			
			116	A	B	B	B	B	B	B	A	A	"	A	B	H	H	H	H	B	L	L	"	"			
			117	A	A	B	B	B	B	B	A	A	"	A	B	L	L	L	L	B	H	H	"	"			
			118	B	B	A	A	A	A	B	A	A	"	B	B	L	L	L	L	B	H	H	"	"			
			119	B	A	A	A	A	A	B	A	A	"	B	B	H	H	H	H	B	H	H	"	"			
			120	B	B	A	A	A	A	B	A	A	"	A	B	H	H	H	H	B	H	H	"	"			
			121	B	A	A	A	A	A	B	A	A	"	A	B	H	H	H	L	B	H	H	"	"			
			122	B	B	A	A	A	A	B	A	A	"	A	B	H	H	H	L	B	H	H	"	"			
			123	B	A	A	A	A	A	B	A	A	"	A	B	H	H	H	H	B	H	H	"	"			
			124	B	B	A	A	A	A	B	A	A	"	A	B	H	H	H	H	B	H	H	"	"			
			125	B	A	A	A	A	A	B	A	A	"	A	B	H	H	H	L	B	H	H	"	"			
			126	B	B	A	A	A	A	B	A	A	"	A	B	H	H	H	L	B	H	H	"	"			
			127	B	A	A	A	A	A	B	A	A	"	A	B	H	H	L	H	B	H	H	"	"			
			128	B	B	A	A	A	A	A	A	A	"	A	B	H	L	H	H	B	H	H	"	"			
			129	B	A	A	A	A	A	A	A	A	"	A	B	H	L	H	H	B	H	H	"	"			
			130	B	B	A	A	A	A	B	A	A	"	A	B	H	L	H	H	B	H	H	"	"			
			131	B	A	A	A	A	A	B	A	A	"	A	B	H	L	H	L	B	H	H	"	"			
			132	B	B	A	A	A	A	B	A	A	"	A	B	H	L	H	L	B	H	H	"	"			
			133	B	A	A	A	A	A	B	A	A	"	A	B	H	L	L	H	B	H	H	"	"			
			134	B	B	A	A	A	A	B	A	A	"	A	B	H	L	L	H	B	H	H	"	"			
			135	B	A	A	A	A	A	B	A	A	"	A	B	H	L	L	L	B	H	H	"	"			
			136	B	B	A	A	A	A	B	A	A	"	A	B	H	L	L	L	B	H	H	"	"			
			137	B	A	A	A	A	A	B	A	A	"	A	B	L	H	H	H	B	H	H	"	"			
			138	B	B	A	A	A	A	B	A	A	"	A	B	L	H	H	H	B	H	H	"	"			
			139	B	A	A	A	A	A	B	A	A	"	A	B	L	H	H	L	B	H	H	"	"			
			140	B	B	A	A	A	A	B	A	A	"	A	B	L	H	H	L	B	H	H	"	"			
			141	B	A	A	A	A	A	B	A	A	"	A	B	L	H	L	H	B	H	H	"	"			
			142	B	B	A	A	A	A	B	A	A	"	A	B	L	H	L	L	B	H	H	"	"			
			143	B	A	A	A	A	A	B	A	A	"	A	B	L	H	L	L	B	H	H	"	"			
			144	B	B	A	A	A	A	B	A	A	"	A	B	L	H	L	L	B	H	H	"	"			
			145	B	A	A	A	A	A	B	A	A	"	A	B	L	L	H	H	B	H	H	"	"			
			146	B	B	A	A	A	A	B	A	A	"	A	B	L	L	H	H	B	H	H	"	"			
			147	B	A	A	A	A	A	B	A	A	"	A	B	L	L	H	L	B	H	H	"	"			
			148	B	B	A	A	A	A	B	A	A	"	A	B	L	L	H	L	B	H	H	"	"			
			149	B	A	A	A	A	A	B	A	A	"	A	B	L	L	L	H	B	H	H	"	"			
			150	B	B	A	A	A	A	B	A	A	"	A	B	L	L	L	H	B	H	H	"	"			
			151	B	A	A	A	A	A	B	A	A	"	A	B	L	L	L	L	B	H	L	"	"			
			152	B	B	A	A	A	A	B	A	A	"	A	B	L	L	L	L	B	L	L	"	"			
			153	B	B	A	A	A	A	B	A	A	"	A	A	L	L	L	L	B	H	H	"	"			
			154	B	A	A	A	A	A	B	A	A	"	A	A	L	L	L	L	B	H	H	"	"			
			155	B	B	A	A	A	A	B	A	A	"	A	B	L	L	L	L	B	L	L	"	"			
			156	B	A	A	A	A	A	B	A	A	"	A	B	L	L	L	H	B	H	H	"	"			
			157	B	B	A	A	A	A	B	A	B	"	A	B	L	L	L	H	B	H	H	"	"			
			158	B	A	A	A	A	A	B	A	B	"	A	B	L	L	L	L	B	H	H	"	"			
			159	B	B	A	A	A	A	B	A	B	"	A	B	L	L	L	L	B	H	H	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																										

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases I,R,S,2 Test no.	1 U/$\overline{D}$	2 CLK	3 A	4 B	5 C	6 D	7 $\overline{ENP}$	8 $\overline{ACLR}$	9 $\overline{SCLR}$	10 GND	11 $\overline{SLOAD}$	12 $\overline{ENT}$	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 $\overline{CCO}$	19 $\overline{RCO}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	f_{MAX} 7/	3003 Fig. 10	160	4.5 V	IN					GND	4.5 V	4.5 V	GND	4.5 V	GND				OUT	GND			5.0 V	QA	30		MHz
	t_{PLH1}		161	"	"	4.5 V				"	"	"	"	GND	"				"	"			"	CLK to QA	4	13	ns
			162	"	"		4.5 V			"	"	"	"	"	"					"			"	CLK to QB	"	"	"
			163	"	"			4.5 V		"	"	"	"	"	"					"			"	CLK to QC	"	"	"
			164	"	"				4.5 V	"	"	"	"	"	"	OUT	OUT	OUT		"			"	CLK to QD	"	"	"
	t_{PHL1}		165	"	"	GND				"	"	"	"	"	"				OUT	"			"	CLK to QA	7	16	"
			166	"	"		GND			"	"	"	"	"	"					"			"	CLK to QB	"	"	"
			167	"	"			GND		"	"	"	"	"	"					"			"	CLK to QC	"	"	"
			168	"	"				GND	"	"	"	"	"	"	OUT	OUT	OUT		"			"	CLK to QD	"	"	"
	t_{PLH2}		169	"	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	"	"					"		OUT	"	CLK to RCO	2	28	"
	t_{PHL2}		170	"	"	GND	GND	GND	GND	"	"	"	"	"	"					"		OUT	"	CLK to RCO	10	19	"
	t_{PLH3}		171	"	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	"	"					"	OUT		"	CLK to CCO	5	13	"
	t_{PHL3}		172	"	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	"	"					"	OUT		"	CLK to CCO	6	25	"
	t_{PLH8}		173	"						"	"	"	"	4.5 V	IN					"		OUT	"	$\overline{ENT}$ to RCO	6	15	"
	t_{PHL8}		174	"						"	"	"	"	"	"					"		OUT	"	$\overline{ENT}$ to RCO	4	13	"
	t_{PLH9}		175	"						"	"	"	"	"	"					"	OUT		"	$\overline{ENT}$ to CCO	5	13	"
	t_{PHL9}		176	"						"	"	"	"	"	"					"	"		"	$\overline{ENT}$ to CCO	9	23	"
	t_{PLH10}		177	"						IN	"	"	"	"	GND					"	"		"	$\overline{ENP}$ to CCO	4	12	"
	t_{PHL10}		178	"						IN	"	"	"	"	"					"	"		"	$\overline{ENP}$ to CCO	5	14	"
	t_{PLH11}		179	IN						GND	"	"	"	"	"					"		OUT	"	U/$\overline{D}$ to RCO	9	23	"
	t_{PHL11}		180	IN						"	"	"	"	"	"					"		OUT	"	U/$\overline{D}$ to RCO	"	19	"
	t_{PHL12}		181	4.5 V	1/	4.5 V				"	IN	"	"	GND	"				OUT	"			"	$\overline{ACLR}$ to QA	"	20	"
			182	"	"		4.5 V			"	"	"	"	"	"					"			"	$\overline{ACLR}$ to QB	"	"	"
			183	"	"			4.5 V		"	"	"	"	"	"		OUT			"			"	$\overline{ACLR}$ to QC	"	"	"
			184	"	"				4.5 V	"	"	"	"	"	"	OUT				"			"	$\overline{ACLR}$ to QD	"	"	"
	t_{PZH}		185	"	"	4.5 V				"	4.5 V	"	"	"	"				OUT	IN			"	$\overline{G}$ to QA	6	18	"
			186	"	"		4.5 V			"	"	"	"	"	"					"			"	$\overline{G}$ to QB	"	"	"
			187	"	"			4.5 V		"	"	"	"	"	"		OUT			"			"	$\overline{G}$ to QC	"	"	"
			188	"	"				4.5 V	"	"	"	"	"	"	OUT				"			"	$\overline{G}$ to QD	"	"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 U/D	2 CLK	3 A	4 B	5 C	6 D	7 $\overline{ENP}$	8 $\overline{ACLR}$	9 $\overline{SCLR}$	10 GND	11 $\overline{SLOAD}$	12 $\overline{ENT}$	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 $\overline{CCO}$	19 $\overline{RCO}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PZL}	3003 Fig. 10	189	4.5 V	1/	GND	GND	GND		GND	4.5 V	4.5 V	GND	GND	GND			OUT	OUT	IN			5.0 V	$\overline{G}$ to QA	11	24	ns
			190	"	"					"	"	"	"	"	"					"			"	$\overline{G}$ to QB	"	"	"
			191	"	"					"	"	"	"	"	"					"			"	$\overline{G}$ to QC	"	"	"
			192	"	"				GND	"	"	"	"	"	"	OUT	OUT			"			"	$\overline{G}$ to QD	"	"	"
	t_{PHZ}		193	"	"	4.5 V				"	"	"	"	"	"					"			"	$\overline{G}$ to QA	1	8	"
			194	"	"		4.5 V			"	"	"	"	"	"					"			"	$\overline{G}$ to QB	"	"	"
			195	"	"			4.5 V		"	"	"	"	"	"					"			"	$\overline{G}$ to QC	"	"	"
			196	"	"				4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{G}$ to QD	"	"	"
	t_{PLZ}		197	"	"	GND				"	"	"	"	"	"					"			"	$\overline{G}$ to QA	3	13	"
			198	"	"		GND			"	"	"	"	"	"					"			"	$\overline{G}$ to QB	"	"	"
			199	"	"			GND		"	"	"	"	"	"					"			"	$\overline{G}$ to QC	"	"	"
			200	"	"				GND	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	$\overline{G}$ to QD	"	"	"
10 T_C = 125°C	f_{MAX}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																							25		MHz
	t_{PLH1}																								4	17	ns
	t_{PHL1}																								7	18	"
	t_{PLH2}																								12	31	"
	t_{PHL2}																								10	22	"
	t_{PLH3}																								5	15	"
	t_{PHL3}																								6	30	"
	t_{PLH8}																								6	17	"
	t_{PHL8}																								4	17	"
	t_{PLH9}																								5	15	"
	t_{PHL9}																								9	28	"
	t_{PLH10}																								4	14	"
	t_{PHL10}																								5	17	"
	t_{PLH11}																								9	25	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 U/D	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 ACLR	9 SCLR	10 GND	11 SLOAD	12 ENT	13 QD	14 QC	15 QB	16 QA	17 G	18 CCO	19 RCO	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10 T_C = +125°C	t_{PHL11}																								9	23	ns
	t_{PHL12}																								9	22	"
	t_{PZH}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C, and limits are as shown.																							6	21	"
	t_{PZL}																								11	29	"
	t_{PHZ}																								1	8	"
	t_{PLZ}																								3	19	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

1/ Apply one clock pulse prior to test as follows: ---2.5 V min/5.5 V max; 0 V.

2/ I_{IL} limits shall be as follows:

Test	Min/max limits in µA for circuit A	B	C
I_{IL}	10/-200		

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.

4/ Only a summary of attributes data is required.

5/ A = 3.0 V minimum; B = 0.0 V or GND.

6/ H >1.5 V; L <1.5 V.

7/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
CCO	Clocked carry output
I_{IN}	Current flowing into an input terminal.
RCO	Ripple carry output
V_{IN}	Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54ALS161B
02	54ALS163B
03	54ALS169
04	54ALS561
05	54ALS569

6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit		
	A	B	C
	Texas Instrument	Motorola Incorporated	National Semi-conductor Corp.
01	X	X	
02	X	X	
03	X		
04	X		
05	X		

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensivenss of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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